

Error Budget in Dispersive Readout

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*A dissertation submitted for the partial fulfilment of BS-MS dual degree in
Science*



Indian Institute of Science Education and Research, Mohali

April 2026

Certificate of Examination

This is to certify that the dissertation titled **Error Budget in Dispersive Readout** submitted by **Ramanuj Raman** (Reg. No. MS21271) for the partial fulfillment of the BS-MS Dual Degree programme of the institute, has been examined by the thesis committee duly appointed by the institute. The committee finds the work done by the candidate satisfactory and recommends that the report be accepted.

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Declaration

The work presented in this dissertation has been carried out by me under the guidance of **Prof. Felix Motzoi** at Forschungszentrum Jülich GmbH, Peter Grünberg Institute, Quantum Control (PGI-8), 52425 Jülich, Germany

This work has not been submitted in part or in full for a degree, a diploma, or a fellowship to any other university or institute. Whenever contributions of others are involved, every effort is made to indicate this clearly, with due acknowledgement of collaborative research and discussions. This thesis is a bonafide record of original work done by me, and all sources listed within have been detailed in the bibliography.

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In my capacity as the supervisor of the candidate's project work, I certify that the above statements by the candidate are true to the best of my knowledge.

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Acknowledgements

I would like to express my sincere gratitude to my supervisor, Prof. Felix Motzoi, for considering me capable of working on this wonderful project. I am also thankful for his invaluable guidance, continuous support, and insightful discussions, as well as his messages on Slack—sometimes late at night—throughout the course of this research.

I am also deeply thankful to Dr. F. A. Cárdenas-López for his guidance, constructive feedback, and support, which greatly contributed to the successful completion of this thesis. I am especially grateful to him for listening to my daily updates and addressing my concerns regarding the project during our bus and train journeys back home, and for consistently providing helpful solutions. These train rides were not only exciting but also highly motivating. I would also like to thank Deutsche Bahn for the occasional delays, which, unexpectedly, allowed for longer and more fruitful discussions—even when I did not have a scheduled meeting.

I extend my appreciation to my administrative guide, Dr. Dipanjan Chakraborty, for his assistance and support in administrative matters. It would not have been possible for me to work on this project without his consent to be my administrative guide, and I would not have been able to enter the IISER campus on my return without his support.

Finally, I would like to thank my friends at PGI-8, IISER Mohali, who helped me through both personal and professional challenges. I am deeply grateful to my parents, Dr. Ram Murti and Rama Kumari, and my brother, Ramashish Naman, for their unwavering support, which enabled me to go to Forschungszentrum Jülich to work on this project.

At last, I would like to dedicate this work to my dear grandmother, Late. Jamuna Devi, for her blessings and encouragement in allowing me to pursue this opportunity. I will always regret that I could not be there with you in your final moments, when you had always been there for me.

Abstract

Dispersive readout is a central technique for measuring superconducting qubits in circuit quantum electrodynamics (cQED), enabling high-fidelity and quantum non-demolition (QND) measurements. In this thesis, we present a detailed theoretical and numerical study of a driven transmon–resonator system operating in the dispersive regime, with the goal of systematically understanding the mechanisms that limit readout performance.

We begin with the full system Hamiltonian describing a coupled qubit-cavity system and derive effective models using unitary transformations, including the Schrieffer–Wolff transformation. By expanding beyond leading order, we capture higher-order corrections that give rise to photon-number-dependent frequency shifts, residual qubit-cavity hybridization, and nonlinear interactions. These corrections play a crucial role in determining the accuracy of dispersive approximations, particularly in regimes of strong drive and finite detuning.

To incorporate dissipation and measurement backaction, we model the system dynamics using a Lindblad master equation, accounting for both coherent evolution and incoherent processes such as Purcell decay and cavity decay. Special attention is given to transformed dissipators in the dispersive frame, which introduce additional photon-dependent decay channels and contribute to the breakdown of QNDness.

We analyze key figures of merit for readout, including signal-to-noise ratio (SNR), readout fidelity, qubit error, and residual cavity population. By comparing effective models with numerically exact simulations, we identify the dominant error mechanisms, including qubit decay, residual photons after measurement, and drive-induced transitions.

Furthermore, we investigate pulse shaping strategies aimed at optimizing readout performance. In particular, we study smooth pulse envelopes and derivative-based corrections to actively suppress residual cavity excitations and improve measurement efficiency. The interplay between pulse design, system parameters (such as coupling strength and detuning), and measurement fidelity is explored in detail.

Overall, this work provides a comprehensive framework for understanding dispersive readout beyond leading-order approximations, offering practical insights into optimizing measurement protocols and mitigating error channels in superconducting quantum devices.

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Chapter 1

Introduction to Circuit Quantum Electrodynamics

Circuit Quantum Electrodynamics(cQED) studies the interaction of artificial atoms based on Josephson junction circuit coupled to quantized field mode encoded in transmission line resonators resembling cavity quantum electrodynamics (CQED) providing a framework for the fundamental study in light matter interaction and its application for development of quantum information devices ranging from different qubits, different readout protocols and amplifiers. In this chapter, we will discuss superconducting LC circuits, superconducting qubits and their interaction with electromagnetic field.

1.1 Superconducting Quantum Circuits

Superconducting quantum circuits corresponds to electronics devices operating in the microwave regime. As the size of the components are a few micrometers and the microwave wavelength is in the millimeter scale they can be treated as lumped elements [Devoret 97] where both voltage and current do not change with the position. In this platform we consider three different lumped elements; capacitors, inductors and Josephson junctions, the later contain a non-linear relation between current and voltage that is helpful to implement non-linear oscillators. In this chapter, we will revisit how to describe a macroscopic circuit as a quantum system, we focus on linear system as the LC circuit and we extend to non-linear oscillators as the Transmon circuit [Koch 07]

1.1.1 Quantum LC Circuit

Consider an LC oscillator with an inductance L and capacitance C connected in series as in Fig(1.1). The circuit have an angular frequency $\omega_r = \frac{1}{\sqrt{LC}}$ and the characteristic impedance

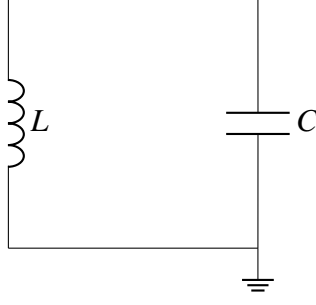


Figure 1.1: Grounded LC resonator circuit

$Z_r = \sqrt{\frac{L}{C}}$. The Hamiltonian of the system is given as

$$H = \frac{Q^2}{2C} + \frac{\Phi^2}{2L} \quad (1.1)$$

where Q is the charge and Φ is the flux in the LC circuit. For this dynamical system, flux (Φ) act as the canonical coordinate and charge (Q) as canonical momentum, hence follow Poisson bracket relation $\{\Phi, Q\} = 1$. Now following the recipe of quantization of a classical Hamiltonian by raising, $Q \rightarrow \hat{Q}$ and $\Phi \rightarrow \hat{\Phi}$ and $\{\Phi, Q\} \rightarrow [\hat{\Phi}, \hat{Q}] = i\hbar$ [Fradkin 13]. In Eq(1.1.1), using ladder operators ($\hat{a}, \hat{a}^\dagger$) to define $\hat{Q} = iQ_0(\hat{a}^\dagger - \hat{a})$ and $\hat{\Phi} = \Phi_0(\hat{a}^\dagger + \hat{a})$ where $\Phi_0 = \sqrt{\frac{\hbar}{2\omega_r C}}$ and $Q_0 = \sqrt{\frac{\hbar\omega_r C}{2}}$ characterise the zero-point fluctuations of the charge and flux. With, these definitions the quantised Hamiltonian is given as

$$\hat{H}_{LC} = \hbar\omega_r \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad (1.2)$$

where $\hat{a}^\dagger \hat{a} |m\rangle = m |m\rangle$ is the number operator representing the number of photons in the resonator, and $|m\rangle$ is the m th photon state represented in the Fock basis

Mathematically, using the above recipe one can quantise any Hamiltonian but the question is can we see the typical properties of a quantum system such as discrete energy levels and presence of superposition (coherence)? To positively answer this question, our system need to satisfy

1. *Ultra-low temperature regime:* Circuits need to be operated at temperatures where the thermal noise does not affect the system avoiding thermal excitation. In this platform such temperature is around the ($\sim 10 - 20mK$) range whose frequency is $k_B T \approx 2\pi \times 1$ GHz whereas for resonators and artificial atoms the frequencies are around $2\pi \times (4 - 7)$ GHz.
2. *High-coherence regime:* We require that the ratio between transition frequencies and losses are high i.e, $Q = \omega_r / \kappa \sim 10^3 - 10^8$ leading that they are sufficiently decoupling from the environment, and the broadening due fluctuations are small allowing to maintain the discrete nature of the system.

By satisfying the above two conditions the LC-circuit can be operated in the quantum regime leading to large coupling between qubit and resonator.

1.1.2 Transmon: An Artificial Atom

The LC-circuit discussed in the earlier section corresponds to our photon source, but to encode an atomic system, we need to add a nonlinearity to the system so that not all the energy levels are accessible with the same transition frequency [Leggett 84]. Thus, as we said before, we need to implement an artificial atom, we need to include Josephson junction to the circuit. In his seminar paper B. Josephson [Josephson 62] demonstrated that it is possible to generate a non-linear current between the two electrodes forming a capacitor if they are close enough and they operate at low temperature leading to the relation

$$I = I_c \sin \varphi \quad (1.3)$$

where φ is the phase difference between two superconducting condensates on the either side of the insulator. The critical current I_c is the maximum amount of current allowed through a Josephson junction leading to dissipation, hence a finite voltage is developed across the junction accompanying a resistive current [Kittel 86]. Josephson also shows a relationship between the Voltage and time derivative of phase difference (φ) as

$$\frac{d\varphi}{dt} = \frac{2\pi}{\Phi_0} V \quad (1.4)$$

where $\Phi_0 = \frac{h}{2e}$ is the magnetic flux quantum. It is useful to relate the phase difference φ with the flux Φ in the circuit via $\Phi(t) = \int^t V(t') dt'$ which will be further used in the discussion. Using Eq(1.1.2), we can define the Josephson Inductance (L_J) as

$$L_J = \left(\frac{dI}{d\Phi} \right)^{-1} = \frac{\Phi_0}{2\pi I_c \cos(2\pi\Phi/\Phi_0)} \quad (1.5)$$

Note that L_J is dependent on Φ unlike the geometric inductance, hence giving a nonlinear character when added to a circuit. Now we can replace the $\frac{\Phi^2}{2L}$ term in Eq (1.1.1) with corresponding term from Josephson junction as $E = \int V(t') I(t') dt' = -E_J \cos\left(\frac{2\pi}{\Phi_0}\Phi\right)$, where $E_J = \frac{\Phi_0 I_c}{2\pi}$ is the Josephson energy. Taking these contributions into account, the quantized Hamiltonian of the capacitively shunted Josephson junction therefore reads

$$\begin{aligned} \hat{H}_T &= \frac{(\hat{Q} - Q_g)^2}{2C_\Sigma} - E_J \cos\left(\frac{2\pi}{\Phi_0}\hat{\Phi}\right) \\ &= 4E_C(\hat{n} - n_g)^2 - E_J \cos \hat{\Phi} \end{aligned} \quad (1.6)$$

It is important to note that in this description the charge degree of freedom is quantized in units of Cooper pairs, such that the relevant charge operator corresponds to $2e$, reflecting

the superconducting nature of the circuit. In Eq (1.6), we have $C_\Sigma = C_J + C_S$ is the total capacitance, including the junction's capacitance C_J and shunted capacitance C_S . Now in the second line, $E_C = \frac{Q^2}{2C_\Sigma}$ is the charging energy, $\hat{n} = \frac{\hat{Q}}{2e}$ is the charge number basis and $\hat{\phi} = \frac{2\pi}{\Phi_0} \hat{\Phi}$ is the phase operator. The offset charge term $n_g = \frac{Q_g}{2e}$ arise from external electric field bias could be because of unwanted degrees of freedom from the environment or an intentional external gate voltage $V_g = Q_g/C_g$. Now the spectrum of $\hat{H}_T$ is controlled by E_C/E_J ratio and for different ratios of E_C/E_J one can realise different types of qubits with different ratios [Blais 21].

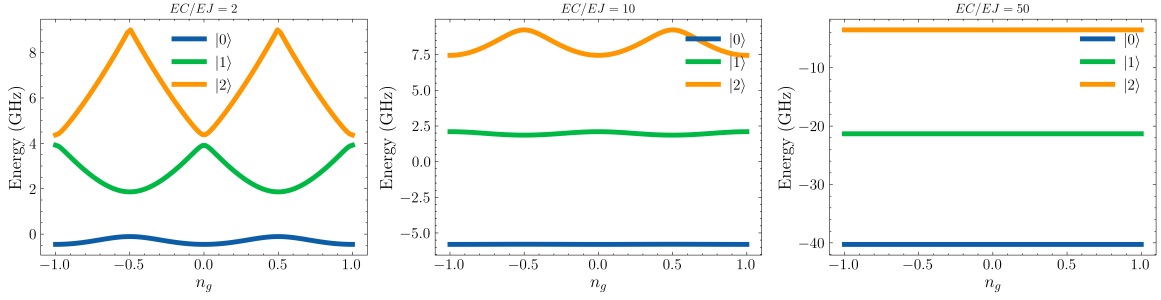


Figure 1.2: Spectrum of Transmon Hamiltonian using numerical diagonalization of Eq (1.6).

In Figure (1.2), one can observe that at large E_C/E_J ratio, the qubit frequency and anharmonicity are insensitive to the charge offset in Eq (1.6). The change in qubit frequency (ω_q) due to unavoidable charge fluctuations in the qubit environment eventually cause dephasing of the qubit. To counter this one works in the transmon regime, where, $E_C/E_J \sim 80 - 90$. With high E_C/E_J , we have high delocalized charge energy due to large josephson energy. While the variance for charge degree of freedom $\Delta n \gg 1$ is large for $E_C/E_J \ll 1$, the variance for its conjugate variable is much smaller $\Delta\phi \ll 1$. Hence we can expand the cosine potential in Eq (1.6) up to fourth order leading

$$\hat{H}_T = 4E_C\hat{n}^2 + E_J \left[\frac{1}{2}\hat{\phi}^2 - \frac{1}{4!}\hat{\phi}^4 \right] \quad (1.7)$$

We have dropped the zeroth order contribution because it is a energy shift that not modify the equation of motion of the system. Following Section 1.1.1 we define the charge ($\hat{n}$) and flux ($\hat{\phi}$) using the creation and annihilation operator as

$$\hat{n} = \frac{i}{2} \left(\frac{E_J}{2E_C} \right)^{1/4} (\hat{b}^\dagger - \hat{b}) \quad (1.8)$$

$$\hat{\phi} = \left(\frac{2E_C}{E_J} \right)^{1/4} (\hat{b}^\dagger + \hat{b}) \quad (1.9)$$

Using Eq (1.8) and Eq (1.9) in Eq (1.1.2) we have the quantized Hamiltonian as

$$\begin{aligned}\hat{H}_T &= \sqrt{8E_CE_J}\hat{b}^\dagger\hat{b} - \frac{E_C}{12}(\hat{b}^\dagger + \hat{b}) \\ &\approx \hbar\omega_q\hat{b}^\dagger\hat{b} - \frac{E_C}{2}\hat{b}^\dagger\hat{b}^\dagger\hat{b}\hat{b}\end{aligned}\quad (1.10)$$

where $\omega_q = (\sqrt{8E_CE_J} - E_C)/\hbar$ is the qubit frequency and the second term is a Kerr nonlinearity, with $E_C/\hbar$ as the frequency shift per oscillation. In principle the above Hamiltonian can be truncated up to two-level system but the correct description of experiments needs multilevel structure of the transmon.

A more accurate description of the system can be provided by accounting higher order terms in the cosine potential by using numerical diagonalization of Eq (1.6) as $\hat{H}_T = \hbar\omega_j|j\rangle\langle j|$ where $|j\rangle$ and $\hbar\omega_j$ corresponds to eigenstates and eigenenergies of the system, respectively. Transmon is one of the most extensively used qubits but many other superconducting qubits can also be realised by using different E_C/E_J ratios. In addition to this other qubits can be designed vary in the number of junctions and topology of the circuit. Various superconducting qubit architectures have been developed, including charge qubits, flux qubits, phase qubits, quantrium, fluxonium, and $0-\pi$ qubits [Nakamura 99, Mooij 99, Martinis 02, Vion 02, Manucharyan 09, Brooks 13].

1.2 Light-Matter Interaction in cQED

In earlier section, we talked about two important components of a superconducting circuit, quantum harmonic oscillator and artificial transmon atom. Now in this section we'll discuss the interaction between the two components which is crucial for qubit control and measurement. Consider a coupled LC transmon circuit in the Figure (1.3).

$$\hat{H} \approx \hbar\omega_r\hat{a}^\dagger\hat{a} + \hbar\omega_q\hat{b}^\dagger\hat{b} - \frac{E_C}{2}\hat{b}^\dagger\hat{b}^\dagger\hat{b}\hat{b} + \hbar g(\hat{b}^\dagger\hat{a} + \hat{b}\hat{a}^\dagger) \quad (1.11)$$

The above Hamiltonian in Eq (1.11) models the interaction of the transmon with an LC oscillator. Reducing the qubit subspace to first two-levels in Eq(1.11) we obtain Jaynes-Cummings Model as

$$\hat{H}_{JC} = \hbar\omega_r\hat{a}^\dagger\hat{a} - \frac{\hbar\omega_q}{2}\hat{\sigma}_z + \hbar g(\hat{a}^\dagger\hat{\sigma}_- + \hat{a}\hat{\sigma}_+) \quad (1.12)$$

where $\hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $\hat{\sigma}_+ = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, and $\hat{\sigma}_- = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

1.2.1 Jaynes Cummings Model

In earlier section we showed that the interaction of transmon with the field can be reduced to a 2-level system and it recovers the Jaynes-Cummings Model. Jaynes Cummings Model

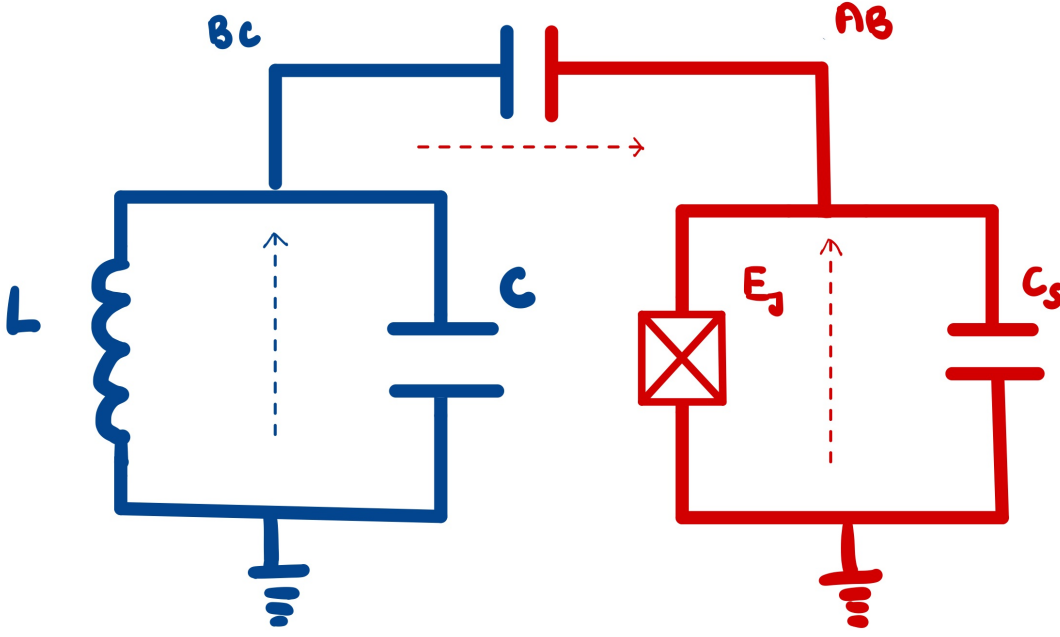


Figure 1.3: A Transmon circuit coupled to a LC-oscillator

is a very important toy model to understand the interaction of the qubit with radiation giving insight into both control and readout of the qubit. In this section we'll discuss the spectrum of the Jaynes Cummings Model. Consider the Eq(1.12) and note that the interaction term links the states such as $|0, n\rangle \rightarrow |1, n-1\rangle$ and write the Hamiltonian as 2×2 matrix.

$$H = \begin{pmatrix} \omega_r n - \omega_q/2 & g\sqrt{n} \\ g\sqrt{n} & \omega_r(n-1) + \omega_q/2 \end{pmatrix} = \omega_r \left(n - \frac{1}{2}\right) \mathbb{I} + \begin{pmatrix} -\Delta/2 & g\sqrt{n} \\ g\sqrt{n} & \Delta/2 \end{pmatrix} \quad (1.13)$$

Now diagonalising the matrix with the off diagonal terms we arrive at the eigenvalues as

$$E_{\pm} = \omega_r n \pm \frac{1}{2} \sqrt{\Delta^2 + 4g^2 n} \quad (1.14)$$

to find the eigenvalues we can use the rotation matrix as

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} -\Delta/2 & g\sqrt{n} \\ g\sqrt{n} & \Delta/2 \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} \omega_r n + \frac{\Delta}{2} \cos 2\theta + g\sqrt{n} \sin 2\theta & g\sqrt{n} \cos 2\theta - \Delta \cos \theta \sin \theta \\ g\sqrt{n} \cos 2\theta - \Delta \cos \theta \sin \theta & \omega_r n - \frac{\Delta}{2} \cos 2\theta - g\sqrt{n} \sin 2\theta \end{pmatrix} \quad (1.15)$$

by setting the off-diagonal terms to zero

$$g\sqrt{n} \cos 2\theta - \Delta \cos \theta \sin \theta = 0 \quad (1.16)$$

hence we have the rotation angle (θ_n) as

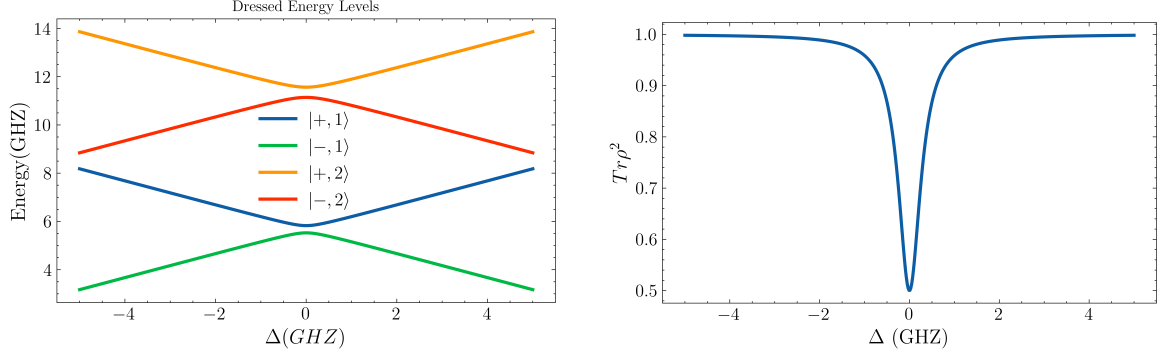
$$\theta_n = \frac{1}{2} \arctan \left(\frac{2g\sqrt{n}}{\Delta} \right) \quad (1.17)$$

Now using rotation matrix and rotation angle from Eq (1.17) we have the eigenvectors as

$$|+,n\rangle = \sin \theta_n |0,n\rangle + \cos \theta_n |1,n-1\rangle \quad (1.18)$$

$$|-,n\rangle = \cos \theta_n |0,n\rangle - \sin \theta_n |1,n-1\rangle \quad (1.19)$$

The energy difference between the two energy levels is $\Delta E = \sqrt{\Delta^2 + 4g^2n}$, for the resonant



(a) Dressed energy levels of the Jaynes-Cummings Hamiltonian

(b) Qubit purity of the first excited state $|- , 1\rangle$

Figure 1.4: Jaynes-Cummings model: (a) dressed energy levels and (b) qubit purity.

condition ($\Delta = 0$), $\Delta E = 2g\sqrt{n}$ as shown in Fig(1.4a). The term $2g\sqrt{n}$, electric dipole coupling, is responsible for lifting the degeneracy of 2-dimensional manifolds of states with n photons.

From Eq(1.14), we can infer that the ground state of the system is $|0,0\rangle$ and is a pure state given the atom and cavity being resonant or not. But for higher levels, for resonant condition ($\Delta = 0$), the two dressed states are

$$|+,n\rangle = \frac{1}{\sqrt{2}} (|0,n\rangle + |1,n-1\rangle) \quad (1.20)$$

$$|-,n\rangle = \frac{1}{\sqrt{2}} (|0,n\rangle - |1,n-1\rangle) \quad (1.21)$$

which show high entanglement between cavity and the qubit, hence the qubit state by itself is not a well defined state. For non resonant ($\Delta \neq 0$) or dispersive limit the dressed states in Eqs(1.18-1.19) can be pure state for $\Delta \rightarrow \infty$ hence $\theta_n \rightarrow 0$, i.e qubit state is a well defined state. In Figure (1.4b), we can see that the qubit purity increase as we increase the detuning between the qubit and cavity. For quantum information processing, it is necessary to work in dispersive regime where the qubit-cavity detuning is much larger with respect to the qubit-cavity coupling: $|g/\Delta| \ll 1$. In this regime, the qubit and cavity are weakly coupled and interaction between the two is not facilitated by the last term in Eq (1.12) but by the virtual photon process, which might be limited in two level system but can be seen in transmon with multi-level structure which will be discussed in later sections.

1.2.2 Beyond Two levels

In earlier section we explored the light-matter interaction of a two-level system in the form of Jaynes-Cummings Model and arrived at the exact solution of the model which is valid in both regimes: resonant and off-resonant. In this section we'll save the multi-level structure of the transmon and study it's interaction with cavity field as given in Eq (1.11).

Consider the transmon Hamiltonian given as

$$\hat{H}_T = \omega_q \hat{b}^\dagger \hat{b} - \frac{E_C}{2} \hat{b}^\dagger \hat{b}^\dagger \hat{b} \hat{b} \quad (1.22)$$

and the eigenenergies of the above Hamiltonian, $\hat{H}_T |j\rangle = E_T(j) |j\rangle$ are given as $E_T(j) = \omega_q + \frac{E_C}{2} - \frac{E_C}{2} j$, hence the eigenenergy of the transmon is decreasing function of the number of excitations. Now to study it's interaction with an LC oscillator we'll reconsider Eq (1.11)

$$\hat{H} \approx \omega_r \hat{a}^\dagger \hat{a} + \omega_q \hat{b}^\dagger \hat{b} - \frac{E_C}{2} \hat{b}^\dagger \hat{b}^\dagger \hat{b} \hat{b} + g(\hat{b}^\dagger \hat{a} + \hat{b} \hat{a}^\dagger) \quad (1.23)$$

the above Hamiltonian can be broken down into a linear part, $\hat{H}_{lin} = \omega_r \hat{a}^\dagger \hat{a} + \hbar \omega_q \hat{b}^\dagger \hat{b} + g(\hat{b}^\dagger \hat{a} + \hat{b} \hat{a}^\dagger)$ and nonlinear part, $\hat{H}_{nlin} = -\frac{E_C}{2} \hat{b}^\dagger \hat{b}^\dagger \hat{b} \hat{b}$. We can diagonalise the linear part of the Hamiltonian using bogoliubov approach. Consider a transformation rule, given as

$$\begin{pmatrix} \hat{a} \\ \hat{b} \end{pmatrix} \rightarrow \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \hat{a} \\ \hat{b} \end{pmatrix} \quad (1.24)$$

Using Eq (1.24) in linear part of the Hamiltonian we have

$$\begin{aligned} \hat{H} = & \frac{1}{2} \left((2g \cos(2\theta) + \sin(2\theta)(\omega_r - \omega_q)) \hat{a}^\dagger \hat{b} + (2g \cos(2\theta) + \sin(2\theta)(\omega_r - \omega_q)) \hat{b}^\dagger \hat{a} \right. \\ & + (2g \sin(2\theta) + \cos(2\theta)(\omega_q - \omega_r) + \omega_q + \omega_r) \hat{b}^\dagger \hat{b} \\ & \left. + (-2g \sin \theta \cos \theta + \omega_q \sin^2 \theta + \omega_r \cos^2 \theta) \hat{a}^\dagger \hat{a} \right) \quad (1.25) \end{aligned}$$

By setting the off-diagonal terms to zero and arriving at the condition $\theta = \frac{1}{2} \arctan\left(\frac{2g}{\Delta}\right)$ where $\Delta = \omega_q - \omega_r$ hence the diagonal Hamiltonian is

$$\hat{H} = \tilde{\omega}_r \hat{a}^\dagger \hat{a} + \tilde{\omega}_q \hat{b}^\dagger \hat{b} \quad (1.26)$$

where $\tilde{\omega}_r = \frac{1}{2} \left(\omega_r + \omega_q - \sqrt{\Delta^2 + 4g^2} \right)$ and $\tilde{\omega}_q = \frac{1}{2} \left(\omega_r + \omega_q + \sqrt{\Delta^2 + 4g^2} \right)$.

Now we can use this similar transformation on the non-linear part of the Hamiltonian as

$$\begin{aligned} \hat{H}_{nl} = & -\frac{E_C}{2} \left(\sin^2(2\theta) \hat{a}^\dagger \hat{b}^\dagger \hat{b} \hat{a} - 2 \sin \theta \cos^3 \theta \hat{a}^\dagger \hat{b}^\dagger \hat{b} \hat{b} \right. \\ & - 2 \sin \theta \cos^3 \theta \hat{b}^\dagger \hat{b}^\dagger \hat{b} \hat{a} + \sin^2 \theta \cos^2 \theta \hat{a}^\dagger \hat{a}^\dagger \hat{b} \hat{b} \\ & + \sin^2 \theta \cos^2 \theta \hat{b}^\dagger \hat{b}^\dagger \hat{a} \hat{a} - 2 \sin^3 \theta \cos \theta \hat{a}^\dagger \hat{a}^\dagger \hat{b} \hat{a} \\ & - 2 \sin^3 \theta \cos \theta \hat{a}^\dagger \hat{b}^\dagger \hat{a} \hat{a} + \sin^4 \theta \hat{a}^\dagger \hat{a}^\dagger \hat{a} \hat{a} \\ & \left. + \cos^4 \theta \hat{b}^\dagger \hat{b}^\dagger \hat{b} \hat{b} \right) \quad (1.27) \end{aligned}$$

Now upto this point everything is exact and now we'll expand the above in a series of λ where $\lambda = \frac{g}{\Delta}$ and then we have $\theta = \frac{1}{2} \arctan\left(\frac{2g}{\Delta}\right)$ now in the small angle approximation we have, $\theta \approx \lambda = \frac{g}{\Delta}$. Hence we can re-write the sine and cosine in Eq (1.27) as

$$\sin \Lambda = \lambda - \frac{\lambda^3}{6} + O(\lambda^5), \quad \cos \Lambda = 1 - \frac{\lambda^2}{2} + O(\lambda^4), \quad (1.28)$$

$$\sin^2 \Lambda = \lambda^2 + O(\lambda^4), \quad \cos^2 \Lambda = 1 - \lambda^2 + O(\lambda^4), \quad (1.29)$$

$$\sin^3 \Lambda = \lambda^3 + O(\lambda^5), \quad \cos^3 \Lambda = 1 - \frac{3}{2}\lambda^2 + O(\lambda^4), \quad (1.30)$$

$$\sin^4 \Lambda = \lambda^4 + O(\lambda^6), \quad \cos^4 \Lambda = 1 - 2\lambda^2 + O(\lambda^4). \quad (1.31)$$

Now using the combinations of above sine and cosine the terms in Eq (1.27) and approximate them upto leading order.

$$-\frac{E_C}{2} \cos^4 \Lambda (\hat{b}^\dagger)^2 \hat{b}^2 \rightarrow -\frac{E_C}{2} (\hat{b}^\dagger)^2 \hat{b}^2, \quad (1.32)$$

$$-\frac{E_C}{2} \sin^4 \Lambda (\hat{a}^\dagger)^2 \hat{a}^2 \rightarrow -\lambda^4 \frac{E_C}{2} (\hat{a}^\dagger)^2 \hat{a}^2, \quad (1.33)$$

$$-2E_C \cos^2 \Lambda \sin^2 \Lambda \hat{a}^\dagger \hat{a} \hat{b}^\dagger \hat{b} \rightarrow -2\lambda^2 E_C \hat{a}^\dagger \hat{a} \hat{b}^\dagger \hat{b}, \quad (1.34)$$

$$E_C \cos^3 \Lambda \sin \Lambda (\hat{b}^\dagger \hat{b} \hat{a}^\dagger \hat{b} + \text{H.c.}) \rightarrow \lambda E_C (\hat{b}^\dagger \hat{b} \hat{a}^\dagger \hat{b} + \text{H.c.}), \quad (1.35)$$

$$E_C \cos \Lambda \sin^3 \Lambda (\hat{a}^\dagger \hat{a} \hat{b}^\dagger + \text{H.c.}) \rightarrow \lambda^3 E_C (\hat{a}^\dagger \hat{a} \hat{b}^\dagger + \text{H.c.}), \quad (1.36)$$

$$-\frac{E_C}{2} \cos^2 \Lambda \sin^2 \Lambda ((\hat{a}^\dagger)^2 \hat{b}^2 + \text{H.c.}) \rightarrow -\lambda^2 \frac{E_C}{2} ((\hat{a}^\dagger)^2 \hat{b}^2 + \text{H.c.}). \quad (1.37)$$

Hence the nonlinear term can be written as

$$\hat{H}'_{\text{nl}} = -\frac{E_C}{2} (\hat{b}^\dagger)^2 \hat{b}^2 - \lambda^4 \frac{E_C}{2} (\hat{a}^\dagger)^2 \hat{a}^2 - 2\lambda^2 E_C \hat{a}^\dagger \hat{a} \hat{b}^\dagger \hat{b} \quad (1.38)$$

$$+ \lambda E_C (\hat{b}^\dagger \hat{b} \hat{a}^\dagger \hat{b} + \text{H.c.}) + \lambda^3 E_C (\hat{a}^\dagger \hat{a} \hat{b}^\dagger + \text{H.c.}) \quad (1.39)$$

$$- \lambda^2 \frac{E_C}{2} ((\hat{a}^\dagger)^2 \hat{b}^2 + \text{H.c.}) + O(\lambda^5). \quad (1.40)$$

Now to get dispersive Hamiltonian we'll use the Schrieffer Wolf Transformation, discussed in more detail in the chapter using the generator given as $\hat{S} = \lambda' (\hat{b}^\dagger \hat{b} \hat{a}^\dagger \hat{b} - \text{H.c.})$ where $\lambda' = \lambda E_C / (\Delta + E_C (1 - 2\lambda^2))$ and neglecting the last two terms we arrive at Hamiltonian given as

$$\hat{H}_{\text{disp}} \approx \tilde{\omega}_r \hat{a}^\dagger \hat{a} + \tilde{\omega}_q \hat{b}^\dagger \hat{b} + \frac{K_a}{2} \hat{a}^\dagger \hat{a}^\dagger \hat{a} \hat{a} + \frac{K_b}{2} \hat{b}^\dagger \hat{b}^\dagger \hat{b} \hat{b} + \chi_{ab} \hat{a}^\dagger \hat{a} \hat{b}^\dagger \hat{a} \quad (1.41)$$

where K_a and K_b are the self-Kerr nonlinearities while third, χ_{ab} is the cross-kerr nonlinearity. These coefficients are defined as

$$K_a = -E_C \left(\frac{g}{\Delta} \right)^4 \quad (1.42)$$

$$K_b = -E_C \quad (1.43)$$

$$\chi_{ab} = -2 \frac{g^2 E_C}{\Delta(\Delta - E_C)} \quad (1.44)$$

which is similar to results in [Koch 07]. Now note that as $E_C \rightarrow 0$ the all nonlinearities vanish. In the dispersive regime all these coefficients are negative but for $0 < \Delta < E_C$ known as straddling regime, is qualitatively different from the dispersive regime, in this regime we observe positive self and cross-nonlinearties ($K_a, K_b, \chi_{ab} > 0$).[Koch 07]

1.2.3 Beyond the transmon: Multilevel Artificial Atom

In the discussion above first we restricted ourselves to a two-level system and realised that the dynamics of the system can be explained using Jaynes Cummings Model and then we considered the transmon-resonator interaction by using the perturbative expansion of the cosine potential under the assumption that $E_J/E_C \gg 1$. To go beyond this regime we'll resort to exact diagonalization of the transmon Hamiltonian in Eq (1.6). Returning back to Hamiltonian given in Eq (A.6) which can be written as

$$\hat{H} = \omega_r \hat{a}^\dagger \hat{a} + 4E_C \hat{n}^2 - E_J \cos \hat{\phi} + 8E_C \hat{n} \hat{n}_r \quad (1.45)$$

Following discussions in [Koch 07], using exact diagonalization of the transmon,

$$\hat{H} = \omega_r \hat{a}^\dagger \hat{a} + \omega_j |j\rangle\langle j| + i g_{ij} |i\rangle\langle j| \left(\hat{a}^\dagger - \hat{a} \right) \quad (1.46)$$

where $|j\rangle$ are energy eigenstates of the bare transmon Hamiltonian and ω_j are the energy eigenvalues of the Hamiltonian obtained from the numerical diagonalization, we also define

$$g_{ij} = 2e \frac{C_g}{CC_\Sigma} Q_{zpf} \langle i | \hat{n} | j \rangle \quad (1.47)$$

The eigenfrequencies ω_j and the matrix element $\langle i | \hat{n} | j \rangle$ is computed numerically in the charge basis, an approach that is also applicable to other superconducting qubits[Blais 21]. Alternatively we can also solve the Eq (A.6) in phase basis, $\hat{\phi} |\varphi\rangle = \varphi |\varphi\rangle$, where Eq (A.6) form of a Mathieu equation whose solution is known.[Koch 07]

Chapter 2

Schrieffer Wolff Transformation

In many quantum systems of practical interest, the full Hamiltonian contains interactions that couple states separated by large energy gaps. Directly solving such Hamiltonians can be difficult, especially when the interaction terms mix different energy manifolds. In such situations, it is often useful to construct an effective Hamiltonian that captures the low-energy dynamics while eliminating rapidly oscillating or off-resonant couplings. The Schrieffer-Wolff transformation (SWT) is a version of degenerate perturbation theory in which the high energy and low energy subspaces are decoupled using a unitary transformation to obtain a low-energy effective hamiltonian[Bravyi 11].

In this chapter we'll discuss the Schrieffer-Wolff Perturbation Theory and its application in readout of superconducting circuits.

2.1 Idea behind the Schrieffer Wolff Transformation

In generic perturbation theory, a Hamiltonian, $\hat{H}$ can be given as sum of two parts: a Hamiltonian with known eigenvalues E_n and eigenfunctions $|\psi_n\rangle$ or the diagonal hamiltonian, $\hat{H}_0$ and the other part is treated as perturbation or off-diagonal hamiltonian, $\lambda \hat{V}$, where λ is the small parameter.

$$\hat{H} = \hat{H}_0 + \lambda \hat{V} \quad (2.1)$$

We assume that we can separate the hilbert space of the system into a desired low-energy subspace P_0 which is invariant under $\hat{H}_0$ and the perturbation $\lambda \hat{V}$ doesn't preserve P_0 . The goal is to construct an effective hamiltonian $\hat{H}_{eff}$ acting only on P_0 such that $\hat{H}_{eff}$ reproduces the eigenvalues of the perturbed hamiltonian $\hat{H} = \hat{H}_0 + \lambda \hat{V}$ originating from P_0 . The Schrieffer Wolff Transformation (SWT) is a unitary operator $\hat{U}$ such that the transformed effective hamiltonian

$$\hat{H}_{eff} = \hat{U}^\dagger (\hat{H}_0 + \lambda \hat{V}) \hat{U} \quad (2.2)$$

preserves P_0 and hence the effective hamiltonian $\hat{H}_{eff}$ is restricted to P_0 . Hence by determining the transformation one can derive an effective hamiltonian.

2.2 Determining the transformation

In this section, we'll determine the transformation to arrive at the effective hamiltonian. Consider $\hat{S}$ as anti-hermitian operator given as $\hat{S}^\dagger = -\hat{S}$, as a generator of SWT, hence $\hat{U} = e^{-\hat{S}}$ is a unitary transformation. Here we assume that

$$\hat{S} = \sum_n \lambda^n \hat{S}^{(n)} \quad (2.3)$$

where the generators upto $\hat{S}^{(n)}$ are needed for an effective hamiltonian of $n+1$ order, $\hat{H}_{eff}^{(n+1)}$. Here we'll determine the generator for each order. By using Baker Campbell Hausdorff (BCH) formula [Arfken 12],

$$e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!} [\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{3!} [\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \dots \quad (2.4)$$

to expand Eq(2.2) with condition $\hat{U} = e^{-\hat{S}}$

$$e^{\hat{S}} \hat{H} e^{-\hat{S}} = H + [\hat{S}, \hat{H}] + \frac{1}{2!} [\hat{S}, [\hat{S}, \hat{H}]] + \frac{1}{3!} [\hat{S}, [\hat{S}, [\hat{S}, \hat{H}]]] + \frac{1}{4!} [\hat{S}, [\hat{S}, [\hat{S}, [\hat{S}, \hat{H}]]]] + \dots \quad (2.5)$$

Now using Eq(2.1) and Eq(2.3) to get a series in λ upto $\mathcal{O}(\lambda^5)$

$$\begin{aligned} e^{\hat{S}} \hat{H} e^{-\hat{S}} &= \hat{H}_0 + \lambda \left([\hat{S}^{(1)}, \hat{H}_0] + \hat{V} \right) \\ &\quad + \lambda^2 \left([\hat{S}^{(2)}, \hat{H}_0] + [\hat{S}^{(1)}, \hat{V}] + \frac{1}{2!} [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{H}_0]] \right) \\ &\quad + \lambda^3 \left([\hat{S}^{(3)}, \hat{H}_0] + [\hat{S}^{(2)}, \hat{V}] + \frac{1}{2!} [\hat{S}^{(1)}, [\hat{S}^{(2)}, \hat{H}_0]] + \frac{1}{2!} [\hat{S}^{(2)}, [\hat{S}^{(1)}, \hat{H}_0]] \right. \\ &\quad \left. + \frac{1}{2!} [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{V}]] + \frac{1}{3!} [\hat{S}^{(1)}, [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{H}_0]]] \right) \\ &\quad + \lambda^4 \left([\hat{S}^{(4)}, \hat{H}_0] + [\hat{S}^{(3)}, \hat{V}] + \frac{1}{2!} [\hat{S}^{(1)}, [\hat{S}^{(3)}, \hat{H}_0]] + \frac{1}{2!} [\hat{S}^{(3)}, [\hat{S}^{(1)}, \hat{H}_0]] + \frac{1}{2!} [\hat{S}^{(1)}, [\hat{S}^{(2)}, \hat{V}]] \right. \\ &\quad + \frac{1}{2!} [\hat{S}^{(2)}, [\hat{S}^{(1)}, \hat{V}]] + \frac{1}{2!} [\hat{S}^{(2)}, [\hat{S}^{(2)}, \hat{H}_0]] + \frac{1}{3!} [\hat{S}^{(1)}, [\hat{S}^{(1)}, [\hat{S}^{(2)}, \hat{H}_0]]] \\ &\quad + \frac{1}{3!} [\hat{S}^{(1)}, [\hat{S}^{(2)}, [\hat{S}^{(1)}, \hat{H}_0]]] + \frac{1}{3!} [\hat{S}^{(2)}, [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{H}_0]]] + \frac{1}{3!} [\hat{S}^{(1)}, [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{V}]]] \\ &\quad \left. + \frac{1}{4!} [\hat{S}^{(1)}, [\hat{S}^{(1)}, [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{H}_0]]]] \right) + \mathcal{O}(\lambda^5) \quad (2.6) \end{aligned}$$

Now we can get an effective hamiltonian upto each order by using the above. Now for $\hat{H}_{eff}^{(2)}$, we set that all other terms $\mathcal{O}(\lambda^n) \rightarrow 0 \forall n \geq 2$ and hence we have the condition

$$[\hat{S}^{(1)}, \hat{H}_0] + \hat{V} = 0 \quad (2.7)$$

Using Eq(2.6), $\hat{S} = \lambda \hat{S}^{(1)}$ and considering terms upto $\mathcal{O}(\lambda^2)$

$$\hat{H}_{eff}^{(2)} = \left([\hat{S}^{(1)}, \hat{V}] + \frac{1}{2!} [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{H}_0]] \right) \quad (2.8)$$

By using condition in Eq(2.7)

$$\hat{H}_{eff}^{(2)} = \left([\hat{S}^{(1)}, \hat{V}] - \frac{1}{2!} [\hat{S}^{(1)}, \hat{V}] \right) = \frac{1}{2!} [\hat{S}^{(1)}, \hat{V}] \quad (2.9)$$

Now we can repeat the above exercise to derive a condition on SWT generator and effective hamiltonian. For $\mathcal{O}(\lambda^2)$, we have

$$[\hat{S}^{(2)}, \hat{H}_0] + [\hat{S}^{(1)}, \hat{V}] + \frac{1}{2!} [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{H}_0]] = 0 \quad (2.10)$$

Again using Eq(2.7),

$$[\hat{S}^{(2)}, \hat{H}_0] + \frac{1}{2!} [\hat{S}^{(1)}, \hat{V}] = 0 \quad (2.11)$$

Here one can note that the last term in Eq(2.9) is there in Eq(2.11) and it would be naive to use the whole structure of term but here it is important to break the term into diagonal and non-diagonal part to prevent singularities, while determining the generator it is important to use just the non-diagonal part of the term. Now we'll use Eq(2.11) to get $\hat{H}_{eff}^{(3)}$

$$\begin{aligned} \hat{H}_{eff}^{(3)} = & [\hat{S}^{(2)}, \hat{V}] + \frac{1}{2!} [\hat{S}^{(1)}, [\hat{S}^{(2)}, \hat{H}_0]] + \frac{1}{2!} [\hat{S}^{(2)}, [\hat{S}^{(1)}, \hat{H}_0]] \\ & + \frac{1}{2!} [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{V}]] + \frac{1}{3!} [\hat{S}^{(1)}, [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{H}_0]]] \end{aligned} \quad (2.12)$$

Using conditions in Eq(2.7) and Eq(2.11)

$$\begin{aligned} \hat{H}_{eff}^{(3)} = & \left(-\frac{1}{4} + \frac{1}{2} - \frac{1}{6} \right) [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{V}]] + \frac{1}{2} [\hat{S}^{(1)}, \hat{V}] \\ = & \frac{1}{2} [\hat{S}^{(2)}, \hat{V}] + \frac{1}{12} [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{V}]] \end{aligned} \quad (2.13)$$

Now for the $\hat{H}_{eff}^{(4)}$ the condition on $\hat{S}^{(3)}$ is given as

$$\begin{aligned} [\hat{S}^{(3)}, \hat{H}_0] + [\hat{S}^{(2)}, \hat{V}] + \frac{1}{2!} [\hat{S}^{(1)}, [\hat{S}^{(2)}, \hat{H}_0]] + \frac{1}{2!} [\hat{S}^{(2)}, [\hat{S}^{(1)}, \hat{H}_0]] \\ + \frac{1}{2!} [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{V}]] + \frac{1}{3!} [\hat{S}^{(1)}, [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{H}_0]]] = 0 \end{aligned} \quad (2.14)$$

hence the $\hat{H}_{eff}^{(4)}$ is given as

$$\begin{aligned} \hat{H}_{eff}^{(4)} = & \frac{1}{2} [\hat{S}^{(3)}, \hat{V}] + \frac{1}{12} [\hat{S}^{(1)}, [\hat{S}^{(2)}, \hat{V}]] + \frac{1}{12} [\hat{S}^{(2)}, [\hat{S}^{(1)}, \hat{V}]] \\ & + \frac{1}{24} [\hat{S}^{(1)}, [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{V}]]] \end{aligned} \quad (2.15)$$

hence the effective hamiltonian is given as

$$\hat{H}_E = \hat{H}_0 + \hat{H}_{eff}^{(2)} + \hat{H}_{eff}^{(3)} + \hat{H}_{eff}^{(4)} \quad (2.16)$$

using this similar method we can arrive at higher order corrections and much broader discussion for higher orders and a time-dependent treatment is discussed in [Malekakhlagh 20]. A similar discussion on the topic can also be found in [Winkler 03]. One should note that blindly following the discussion above may lead to some singularities in the expression which can be avoided at each step by separating the off-diagonal and diagonal terms in the effective hamiltonian and using the off-diagonal terms to guess an ansatz which to be determined using conditions discussed above. For our purpose fourth-order corrections are enough and in the next section we'll discuss the limits of SWT.

2.2.1 Limits of SWT

As mentioned in earlier in 2.1 that SWT is a quasi-degenerate perturbation theory hence only valid when $\hat{V}$ is sufficiently small. While working in dispersive regime it is practical from experimental point of view as well from a physics point of view to understand the limits of the SWT. Consider the a two level system evolving under the following hamiltonian,

$$\hat{H} = \begin{pmatrix} E_0 & \lambda V \\ \lambda V & E_1 \end{pmatrix} \quad (2.17)$$

the eigenvalues of the above hamiltonian are

$$E_{\pm} = \frac{(E_0 + E_1) \pm \sqrt{(E_0 - E_1)^2 + 4\lambda^2 V^2}}{2} \quad (2.18)$$

to understand the radius of convergence of the perturbation theory, in Eq(2.18) one encounters a branch point at

$$\lambda|V| = \frac{\pm i(E_0 - E_1)}{2}$$

hence the condition for the convergence of the perturbation series is

$$|V| < \frac{\Delta}{2} \quad (2.19)$$

where $\Delta = |E_0 - E_1|$ [Sakurai 17].

The above analysis motivates for a condition for convergence of SWT series as given in [Bravyi 11]

$$2\|\hat{V}\| < \Delta_{min} \quad (2.20)$$

2.3 Jaynes Cummings Model in Dispersive Regime

Due to the non-resonant condition ($\Delta \neq 0$), and $|g/\Delta| \ll 1$ Jaynes Cummings model Eq(1.12) can be transformed to an effective model using second-order perturbation theory. Consider Eq(1.12) in which we have two parts one that is system hamiltonian and the second part as interaction. Here the system hamiltonian consist of qubit hamiltonian and the resonator hamiltonian, $\hat{H}_0 = \omega_r \hat{a}^\dagger \hat{a} - \frac{\omega_q}{2} \hat{\sigma}_z$ and the interaction term is, $\hat{V} = g (\hat{a}^\dagger \hat{\sigma}_- + \hat{a} \hat{\sigma}_+)$. Here we can use Schrieffer Wolf Transformation(SWT) to get an effective hamiltonian in second order of perturbation, the effective hamiltonian is given as

$$\hat{H}_{eff}^{(2)} = \hat{H}_0 + \frac{1}{2} [\hat{S}^{(1)}, \hat{V}] \quad (2.21)$$

where

$$\hat{S}^{(1)} = \frac{g}{\Delta} (\hat{a}^\dagger \hat{\sigma}_- - \hat{a} \hat{\sigma}_+) \quad (2.22)$$

For the second term in the Eq(2.21)

$$\begin{aligned} [\hat{S}^{(1)}, \hat{V}] &= \left[\frac{g}{\Delta} (\hat{a}^\dagger \hat{\sigma}_- - \hat{a} \hat{\sigma}_+), g (\hat{a}^\dagger \hat{\sigma}_- + \hat{a} \hat{\sigma}_+) \right] \\ &= \frac{2g^2}{\Delta} [\hat{a}^\dagger \hat{\sigma}_-, \hat{a} \hat{\sigma}_+] \end{aligned} \quad (2.23)$$

Using relation, $[AB, CD] = AC[B, D] + [A, C]DB$ [Sakurai 17]

$$[\hat{a}^\dagger \hat{\sigma}_-, \hat{a} \hat{\sigma}_+] = \hat{a}^\dagger \hat{a} [\hat{\sigma}_-, \hat{\sigma}_+] + [\hat{a}^\dagger, \hat{a}] \hat{\sigma}_+ \hat{\sigma}_- \quad (2.24)$$

Using $[\hat{a}^\dagger, \hat{a}] = -1$, $[\hat{\sigma}_-, \hat{\sigma}_+] = \hat{\sigma}_z$ and $\hat{\sigma}_+ \hat{\sigma}_- = |1\rangle\langle 1| = \frac{1}{2}(1 - \hat{\sigma}_z)$

$$[\hat{a}^\dagger \hat{\sigma}_-, \hat{a} \hat{\sigma}_+] = \hat{a}^\dagger \hat{a} \hat{\sigma}_z + \frac{\hat{\sigma}_z}{2} - \frac{1}{2} \quad (2.25)$$

Using Eq(2.25) in Eq(2.23) we have the effective hamiltonian

$$\begin{aligned} \hat{H}_{eff}^{(2)} &= \omega_r \hat{a}^\dagger \hat{a} + \frac{g^2}{\Delta} \hat{a}^\dagger \hat{a} \hat{\sigma}_z + \frac{g^2}{2\Delta} \hat{\sigma}_z - \frac{g^2}{2\Delta} - \frac{\omega_q}{2} \hat{\sigma}_z \\ &= \left(\omega_r + \frac{g^2}{\Delta} \hat{\sigma}_z \right) \hat{a}^\dagger \hat{a} - \frac{1}{2} \left(\omega_q - \frac{g^2}{\Delta} \right) \hat{\sigma}_z - \frac{g^2}{2\Delta} \mathbb{I} \end{aligned} \quad (2.26)$$

we can neglect the constant energy shift in the hamiltonian $-\frac{g^2}{2\Delta} \mathbb{I}$ and the effective hamiltonian is

$$\hat{H}_{eff}^{(2)} = \left(\omega_r + \frac{g^2}{\Delta} \hat{\sigma}_z \right) \hat{a}^\dagger \hat{a} - \frac{1}{2} \left(\omega_q - \frac{g^2}{\Delta} \right) \hat{\sigma}_z \quad (2.27)$$

Now note the new terms in the effective hamiltonian, there is a shift in the cavity frequency as $\left(\omega_r + \frac{g^2}{\Delta} \hat{\sigma}_z \right)$ and the shift in the cavity frequency is known as AC stark shift or dispersive shift. This dispersive shift is exploited for the qubit readout. There is a similar term in the qubit frequency as $\left(\omega_q - \frac{g^2}{\Delta} \right)$ which shifts the qubit frequency by $\frac{g^2}{\Delta}$ and known as lamb

shift.

Employing the discussion in Sec 2.2 to get the $\mathcal{O}(\lambda^4)$ correction we need to determine $\hat{S}^{(2)}$ and $\hat{S}^{(3)}$. From Eq (2.11), the term $[\hat{S}^{(1)}, \hat{V}]$ is diagonal hence $\hat{S}^{(2)} = 0$. So the condition for $\hat{S}^{(3)}$ is given as

$$[\hat{S}^{(3)}, \hat{H}_0] + \frac{1}{2!} [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{V}]] + \frac{1}{3!} [\hat{S}^{(1)}, [\hat{S}^{(1)}, [\hat{S}^{(1)}, \hat{H}_0]]] = 0 \quad (2.28)$$

Using Eq(2.22) and Eq(1.12) in Eq(2.28)

$$\hat{S}^{(3)} = \frac{g^3}{3\Delta^3} (\hat{a}^\dagger \hat{\sigma}_- - \hat{a} \hat{\sigma}_+) + \frac{g^3}{3\Delta^3} (\hat{a}^\dagger \hat{a}^\dagger \hat{a} \hat{\sigma}_- - \hat{a}^\dagger \hat{a} \hat{a} \hat{\sigma}_+) \quad (2.29)$$

and using $\hat{S}^{(3)}$, $\hat{S}^{(2)}$ and $\hat{S}^{(1)}$ in Eq(2.15) and the expression for $\hat{H}_{eff}^{(4)}$ is

$$\hat{H}_{eff}^{(4)} = -\frac{g^4}{3\Delta^3} - \frac{2g^4}{3\Delta^3} a^\dagger a - \frac{4g^4}{3\Delta^3} a^\dagger a \sigma_z - \frac{2g^4}{3\Delta^3} a^\dagger a^\dagger a a \sigma_z - \frac{g^4}{3\Delta^3} \sigma_z \quad (2.30)$$

hence the effective hamiltonian upto $\mathcal{O}(\frac{g^4}{\Delta^3})$ is given as

$$\begin{aligned} \hat{H}_{eff} = & \left(\omega_r - \frac{2g^4}{3\Delta^3} \right) a^\dagger a + \left(-\frac{\omega_q}{2} + \frac{g^2}{2\Delta} - \frac{g^4}{3\Delta^3} \right) \sigma_z \\ & + \left(\frac{g^2}{\Delta} - \frac{4g^4}{3\Delta^3} \right) a^\dagger a \sigma_z - \frac{2g^4}{3\Delta^3} a^\dagger a^\dagger a a \sigma_z + \left(\frac{g^2}{2\Delta} - \frac{g^4}{3\Delta^3} \right) \end{aligned} \quad (2.31)$$

Now here we see that there are terms in addition to the second order effect such as independent of qubit state frequency shift in the cavity frequency and an additional shift fourth-order frequency shift in the qubit. Apart from the dispersive shift which appear as a second order effect there is also a fourth-order term which change the cavity frequency depending upon the state of the qubit, with a nonlinear dependence upon the cavity photons, this effect is called nonlinear Kerr-effect, which describes the change in the refractive index of the material proportional to the square of the applied electric field.

In this section we saw an application of the Schrieffer Wolf Perturbation Theory and arrived at an effective description of the system but as discussed in Section 2.2.1 there is a limit beyond which the approximation breaks and the effective model is no longer useful to describe the physics of the system. Now to see what Eq(2.20) means in the context of Jaynes Cummings model we need to first evaluate $||\hat{V}||$, which is defined as $||\hat{V}|| = \langle n, j | \hat{V}^\dagger \hat{V} | n, j \rangle^{1/2}$ where n is number of the fock states and j is the level of the qubit [Blais 21]. In Jaynes Cummings Hamiltonian,

$$||\hat{V}|| = g \langle n, j | (\hat{a}^\dagger \sigma_- + \hat{a} \sigma_+) (\hat{a}^\dagger \sigma_- + \hat{a} \sigma_+) | n, j \rangle^{1/2} = 2g\sqrt{n} \quad (2.32)$$

and here $\Delta_{min} = \Delta$ using above equations in Eq(2.20)

$$n_{crit} = \left(\frac{\Delta}{2g} \right)^2 \quad (2.33)$$

hence for $\bar{n} = n_{crit}$ the dispersive approximation breaks and the effective model is no longer describes the right physics of the system.

Chapter 3

Readout of Superconducting Qubits

In this chapter, we discuss the mechanisms for qubit readout, with particular focus on dispersive readout, which is widely used across many superconducting qubit architectures[Blais 07, Blais 21]. Dispersive readout is realized by measuring the scattering of a probe tone off an oscillator coupled to a qubit. Advantages of such approach is that the probe can be turned off hence reduced measurement induced backaction. In earlier chapters we have discussed the Jaynes-Cummings Model in dispersive regime and a transmon coupled to a LC oscillator. In this chapter we will discuss the driven-dissipative dynamics of a transmon coupled to a resonator driven at off-resonance and how we take a readout of a superconducting qubit?

3.1 Driven-Dissipative Transmon

Consider a transmon coupled to a readout cavity driven at a frequency (ω_d), off resonant to the qubit ($\omega_r \neq \omega_q$). The Hamiltonian of the system includes two terms, the time-independent transmon-oscillator Hamiltonian and the resonator drive.

$$\hat{H} = \omega_r \hat{a}^\dagger \hat{a} + \epsilon_i |j\rangle\langle j| + g_{jk}(\hat{a}^\dagger |j\rangle\langle k| + \hat{a} |k\rangle\langle j|) + \Omega(t) \cos \omega_d t (\hat{a}^\dagger + \hat{a}) \quad (3.1)$$

Now we move to a rotating frame to cancel fast rotating terms. Let $\hat{U} = \exp(-i\omega_d(\hat{a}^\dagger \hat{a} + n|n\rangle\langle n|))$ and the effective hamiltonian ($\hat{H}_{eff}$) is

$$\hat{H}_{eff} = \hat{U}^\dagger \left(i \frac{\partial}{\partial t} - \hat{H} \right) \hat{U} = i\hat{U}^\dagger \dot{\hat{U}} - \hat{U}^\dagger \hat{H} \hat{U} \quad (3.2)$$

Now $i\hat{U}^\dagger \dot{\hat{U}} = \omega_d \hat{a}^\dagger \hat{a} + \omega_d n |n\rangle\langle n|$. For the second term, using Eq(2.4)

1.

$$\hat{U}^\dagger |j\rangle\langle k| \hat{U} = |j\rangle\langle k| \exp(i\omega_d(j-k)t) \quad (3.3)$$

2.

$$\hat{U}^\dagger \hat{a} \hat{U} = \hat{a} - i\omega_d \hat{a} - \frac{\omega_d^2}{2!} \hat{a} + \mathcal{O}(\omega_d^3) = \hat{a} \exp(-i\omega_d t) \quad (3.4)$$

3.

$$\hat{U}^\dagger \hat{a}^\dagger \hat{a} \hat{U} = \hat{a}^\dagger \hat{a} + i\omega_d [\hat{a}^\dagger \hat{a}, \hat{a}^\dagger \hat{a}] + \mathcal{O}(\omega_d^2) = \hat{a}^\dagger \hat{a} \quad (3.5)$$

All the commutators vanish for order 1 or higher, $[\hat{a}^\dagger \hat{a}, \hat{a}^\dagger \hat{a}] = [\hat{a}^\dagger \hat{a}, [\hat{a}^\dagger \hat{a}, \hat{a}^\dagger \hat{a}]] = 0$

Using Eqs(3.3-3.5) in $\hat{U}^\dagger \hat{H} \hat{U}$ so we have

$$\begin{aligned} \hat{U}^\dagger \hat{H} \hat{U} = & \omega_r \hat{U}^\dagger \hat{a}^\dagger \hat{a} \hat{U} + \varepsilon_j \hat{U}^\dagger |j\rangle\langle j| \hat{U} + g_{jk} \left(\hat{U}^\dagger \hat{a}^\dagger |j\rangle\langle k| \hat{U} + \hat{U}^\dagger \hat{a} |k\rangle\langle j| \hat{U} \right) \\ & + \Omega(t) \cos \omega_d t (\hat{U}^\dagger \hat{a}^\dagger \hat{U} + \hat{U}^\dagger \hat{a} \hat{U}) \end{aligned} \quad (3.6)$$

Now for the first and second term in the above equation from Eq(3.5) and Eq(3.3) remain the same. For third term,

$$\hat{U}^\dagger \hat{a}^\dagger |j\rangle\langle k| \hat{U} = \hat{a} |j\rangle\langle k| \exp(i\omega_d(j-k-1)t) \quad (3.7)$$

Now for the drive term we have,

$$\Omega(t) \left(\frac{e^{i\omega_d t} + e^{-i\omega_d t}}{2} \right) \hat{U}^\dagger (\hat{a}^\dagger + \hat{a}) \hat{U} = \frac{\Omega(t)}{2} \left(\hat{a}^\dagger \exp(i2\omega_d t) + \hat{a} \exp(-i2\omega_d t) + \hat{a}^\dagger + \hat{a} \right) \quad (3.8)$$

Now here we'll drop the rotating term in the drive and the qubit-photon interaction, in Eq(3.7) for non-rotating term, $j = k + 1$. Hence the effective hamiltonian, using Eq(3.2) is

$$\hat{H}_{eff} = \Delta \hat{a}^\dagger \hat{a} + e_k |k\rangle\langle k| + g_{k,k+1} \left(\hat{a} |k\rangle\langle k+1| + \hat{a}^\dagger |k+1\rangle\langle k| \right) + \frac{\Omega(t)}{2} (\hat{a}^\dagger + \hat{a}) \quad (3.9)$$

where $\Delta = \omega_r - \omega_d$ and $e_k = \varepsilon_k - k\omega_d$. We'll use $\Omega(t)/2 \rightarrow \Omega(t)$ and $\hat{H}_{eff} \rightarrow \hat{H}$ in the further discussion.

Hence the new hamiltonian in the rotating frame is

$$\hat{H} = \Delta \hat{a}^\dagger \hat{a} + e_k |k\rangle\langle k| + g_{k,k+1} \left(\hat{a} |k\rangle\langle k+1| + \hat{a}^\dagger |k+1\rangle\langle k| \right) + \frac{\Omega(t)}{2} (\hat{a}^\dagger + \hat{a}) \quad (3.10)$$

3.1.1 Dissipation in the system

Since the system is not only driven but also dissipates through channels such as qubit relaxation, qubit dephasing and resonator decay. Under the well established Born-Markov approximations, following Lindblad-form Markovian master equations for the system's density matrix ρ [Breuer 02]

$$\dot{\rho} = -i[\hat{H}, \rho] + \kappa \mathcal{D}[\hat{a}] \rho + \gamma \mathcal{D}[\hat{b}] \rho + 2\gamma_\phi \mathcal{D}[\hat{b}^\dagger \hat{b}] \rho \quad (3.11)$$

where $\hat{H}$ is defined in Eq(3.10), $\mathcal{D}[\hat{O}] = \hat{O} \hat{O}^\dagger - \frac{1}{2} \{ \hat{O}^\dagger \hat{O}, \}$, κ is the decay rate of the resonator, γ is the qubit decay rate and γ_ϕ is the qubit dephasing rate. These rates γ and γ_ϕ

appearing in the previous expression are related to the characteristic T_1 relaxation time and T_2 coherence time of the qubit, which are defined as[Schoelkopf 03]

$$T_1 = \frac{1}{\gamma_1} = \frac{1}{\gamma_\uparrow + \gamma_\downarrow} \simeq \frac{1}{\gamma} \quad (3.12)$$

and

$$T_2 = \frac{1}{\gamma_2} = \left(\frac{\gamma_1}{2} + \gamma_\phi \right)^{-1} \quad (3.13)$$

At zero temperature, T_1 is the characteristic time in which an artificial atom decays from excited state to ground state and T_2 is the characteristic lifetime of the coherent superposition, which includes a contribution from both dephasing and relaxation.

In phenomenological description of qubit dephasing, γ_ϕ can be attributed to charge dispersion and from Figure(1.2) the quantity is small for $0 - 1$ transition but it increases exponentially as we move up the transmon levels[Egger 19]. In addition to that the residual photons in the cavity shifts the qubit frequency due to dispersive shift $\chi_{ab}\hat{a}^\dagger\hat{a}\hat{b}^\dagger\hat{b}$ as shown in Eq(1.41). One can also re-write the above equation for a transmon in a two-level approximation by replacing $\hat{b} \rightarrow \sigma_-$, $\hat{b}^\dagger \rightarrow \sigma_+$, $\hat{b}^\dagger\hat{b} \rightarrow (\sigma_z + 1)/2$ and the qubit hamiltonian is given as $\hat{H}_q \rightarrow \hbar\omega_q\sigma_z/2$ hence Eq(3.11) can be re-written as

$$\dot{\rho} = -i[\hat{H}_q, \rho] + \kappa\mathcal{D}[\hat{a}]\rho + \gamma\mathcal{D}[\sigma_-]\rho + \frac{\gamma_\phi}{2}\mathcal{D}[\sigma_z]\rho \quad (3.14)$$

For the current state of the art one can achieve T_1 upto $50-120\mu s$ [Devoret 13, Kjaergaard 20, Wei 20] some reports are also there for T_1 upto $300\mu s$ using different materials for the fabrication[Place 21]. Other superconducting qubits also shows large relaxation and coherence times such as heavy-fluxonium qubits have $T_1, T_2 \sim 300\mu s$ [Zhang 21] and for $0 - \pi$ qubits $T_1 \sim 1.6ms$ and $T_2 \sim 25\mu s$ [Gyenis 21]. Current willow processor by Google have a relaxation time of about $98\mu s$. [Google Quantum AI 24]

3.2 Transmon-Cavity Interaction in Dispersive Regime

In the last section we saw that for the readout of a superconducting apart from the Jaynes-Cummings type interaction between the transmon and cavity there is a drive on the cavity and the cavity dissipates at a rate κ . Since we are driving the system at an off-resonant frequency we can use Schrieffer Wolf Transformation to describe the dynamics using an effective hamiltonian as discussed in Chapter 2. In this section we'll repeat the same exercise as shown in Section 2.3 but utilising the multilevel structure of the transmon and considering the cosine potential without any perturbative expansion.

Consider the hamiltonian in Eq (3.10) where

$$\hat{H}_0 = \Delta\hat{a}^\dagger\hat{a} + e_k|k\rangle\langle k|$$

and

$$\hat{V} = g_{k,k+1} \left(\hat{a}^\dagger |k\rangle\langle k+1| + \hat{a} |k+1\rangle\langle k| \right)$$

Using $\hat{V}$ and $\hat{H}_0$ in Eq(2.7) to calculate $\hat{S}^{(1)}$,

$$\hat{S}^{(1)} = \frac{g_{k,k+1}}{(\omega_r + \epsilon_k - \epsilon_{k+1})} \left(\hat{a}^\dagger |k\rangle\langle k+1| - \hat{a} |k+1\rangle\langle k| \right) \quad (3.15)$$

where ω_r is the resonator frequency, $g_{k,k+1}$ is the interaction between levels $k \rightarrow k+1$ and ϵ_k are the bare energies of the transmon. Now using Eq (3.15) in Eq (2.21)

$$\begin{aligned} \hat{H}_{eff}^{(2)} = & \left(\frac{g_{k,k+1}^2}{(\omega_r + \epsilon_k - \epsilon_{k+1})} - \frac{g_{k-1,k}^2}{(\omega_r + \epsilon_{k-1} - \epsilon_k)} \right) |k\rangle\langle k| \hat{a}^\dagger \hat{a} - \frac{g_{k,k+1}^2}{(\omega_r + \epsilon_k - \epsilon_{k+1})} |k+1\rangle\langle k+1| \\ & + \frac{1}{2} \left(\frac{g_{k,k+1}g_{k+1,k+2}}{(\omega_r + \epsilon_k - \epsilon_{k+1})} - \frac{g_{k,k+1}g_{k+1,k+2}}{(\omega_r + \epsilon_{k+1} - \epsilon_{k+2})} \right) \hat{a} \hat{a} |k\rangle\langle k+2| + h.c. \end{aligned} \quad (3.16)$$

the first term in the hamiltonian is the dispersive shift, second term is the lamb shift and the last term is an off-diagonal term. Note that there was no such off-diagonal term observed in two level system given in Eq (2.27) and these off-diagonal terms also vanish in two level approximation in the above equation.

Now using the off-diagonal term in Eq (2.14) to compute $\hat{S}^{(3)}$ we have

$$\hat{S}^{(2)} = \frac{(\epsilon_k - 2\epsilon_{k+1} + \epsilon_{k+2})g_k g_{k+1}}{2(\omega_r + \epsilon_k - \epsilon_{k+1})(2\omega_r + \epsilon_k - \epsilon_{k+2})(\omega_r + \epsilon_{k+1} - \epsilon_{k+2})} \hat{a} \hat{a} |k\rangle\langle k+2| - h.c. \quad (3.17)$$

Now using Eq (2.13) to arrive at the third order hamiltonian and one realise there is no diagonal contribution and only off-diagonal terms

$$\begin{aligned} \hat{H}_{eff}^{(3)} = & \left[C_k^{(1)} \left(\hat{a} |k+1\rangle\langle k| + \hat{a}^\dagger |k\rangle\langle k+1| \right) \right. \\ & + C_k^{(2)} \left(\hat{a} \hat{a} |k+3\rangle\langle k| + \hat{a}^\dagger \hat{a}^\dagger |k\rangle\langle k+3| \right) \\ & \left. + C_k^{(3)} \left(\hat{a}^\dagger \hat{a} \hat{a} |k+1\rangle\langle k| + \hat{a}^\dagger \hat{a}^\dagger \hat{a} |k\rangle\langle k+1| \right) \right] \end{aligned} \quad (3.18)$$

Now note that all these terms are off-diagonal and make no contribution to energy shifts in the diagonal. Hence we can collect all the off-diagonal components and get $\hat{S}^{(3)}$ using Eq(2.14):

$$\begin{aligned} \hat{S}^{(3)} = & C_k^{(1)} \left(\hat{a} |k+1\rangle\langle k| - \hat{a}^\dagger |k\rangle\langle k+1| \right) \\ & + C_k^{(2)} \left(\hat{a}^\dagger \hat{a} \hat{a} |k+1\rangle\langle k| - \hat{a}^\dagger \hat{a}^\dagger \hat{a} |k\rangle\langle k+1| \right) \\ & + C_k^{(3)} \left(\hat{a} \hat{a} \hat{a} |k+3\rangle\langle k| - \hat{a}^\dagger \hat{a}^\dagger \hat{a}^\dagger |k\rangle\langle k+3| \right) \end{aligned} \quad (3.19)$$

Now we can use Eq(3.15), Eq(3.19), and Eq(3.17) in Eq(2.15) to obtain the effective Hamiltonian for the fourth order and sum them to arrive at an effective hamiltonian

$$\hat{H}_{eff} = (\omega_r - \omega_d) \hat{a}^\dagger \hat{a} + \tilde{\epsilon}_k |k\rangle\langle k| + \chi_k \hat{a}^\dagger \hat{a} |k\rangle\langle k| + K_k \hat{a}^\dagger \hat{a} \hat{a} |k\rangle\langle k| \quad (3.20)$$

where

$$\tilde{\epsilon}_k = (\epsilon_k - k\omega_d) - \frac{g_{k-1,k}^2}{\omega_r + \epsilon_{k-1} - \epsilon_k} + \frac{g_{k-1,k}^2 P_k^{(L)}}{6(\omega_r + \epsilon_{k-2} - \epsilon_{k-1})^2 (2\omega_r + \epsilon_{k-2} - \epsilon_k) (\omega_r + \epsilon_{k-1} - \epsilon_k)^3}, \quad (3.21)$$

$$\chi_k = \frac{g_{k,k+1}^2}{\omega_r + \epsilon_k - \epsilon_{k+1}} - \frac{g_{k-1,k}^2}{\omega_r + \epsilon_{k-1} - \epsilon_k} + \frac{P_k^{(D)}}{3D_k}, \quad (3.22)$$

$$K_k = -\frac{P_k^{(K)}}{12D_k} \quad (3.23)$$

Where the shared 4th-order denominator is defined as:

$$D_k = (\omega_r + \epsilon_{k-2} - \epsilon_{k-1})^2 (2\omega_r + \epsilon_{k-2} - \epsilon_k) (\omega_r + \epsilon_{k-1} - \epsilon_k)^3 (\omega_r + \epsilon_k - \epsilon_{k+1})^3 \quad (3.24)$$

Note: The explicit definitions for the transformation coefficients (C_k) and the polynomial numerators (P_k) are deferred to the Appendix B

Here χ_k and K_k are the state dependent energy shift of second and fourth order respectively. In Eq(3.21) there are terms from higher order corrections too along with the lamb shift as discussed in Section 2.3.

3.2.1 Transformed Drive and Dissipator

Now as we discussed in Chapter 2, SWT is a unitary transformation and the aim in the earlier exercise has been to arrive at a block diagonal structure for the hamiltonian. Using SWT we have caused a frame transformation and now we'll use that frame transformation to see what other terms such as drive and dissipator looks like which give us insight into what kind of process are happening in the system at different orders. Consider the BCH expansion given in Eq(2.6) replace $\hat{H}$ with $\hat{a}$ and then we can a similar expansion in λ . Now in the dressed frame the cavity annihilation operator transforms upto $\mathcal{O}(\lambda^2)$ as

$$\hat{a} \rightarrow \hat{a} - \frac{g_{k,k+1}}{(e_k - e_{k+1} + \omega_r)} |n\rangle\langle n+1| \quad (3.25)$$

In the transformed frame, cavity annihilation operator have an additional term which cause transition from $|k+1\rangle \rightarrow |k\rangle$, a similar result can be arrived for a two-level system where using the first order Schriffer Wolf generator where, $\hat{a} \rightarrow \hat{a} + \frac{g}{\Delta} \sigma_-$. This phenomena is known as the purcell decay[Purcell 46]. In the dressed frame, the qubit dressed by the field can relax through cavity via releasing a photon at a rate of $\kappa \left(\frac{g}{\Delta}\right)^2$ where κ is the rate of cavity decay.

Similarly in the dressed frame the resonator drive transforms as

$$\Omega(t) (\hat{a}^\dagger + \hat{a}) \rightarrow \Omega(t) (\hat{a}^\dagger + \hat{a}) - \Omega(t) \frac{g_{k,k+1}}{(e_k - e_{k+1} + \omega_r)} (|k\rangle\langle k+1| + |k+1\rangle\langle k|) \quad (3.26)$$

in a two level approximation the last term in the above equation can be written as $\Omega(t) \frac{g}{\Delta} \sigma_x$ where the drive can have both real and imaginary drive. From this analysis we understand two significant mechanisms which affect the transmon during readout are Purcell decay and qubit driving at first order. Now we'll extend this to higher orders and examine different processes that appear in the drive and the dissipator.

The terms that appear at $\mathcal{O}(\lambda^2)$ are

$$\begin{aligned} \hat{a}^{(2)} = & \frac{1}{2} \sum_k \frac{g_{k,k+1}^2}{(\omega_r + e_k - e_{k+1})^2} \hat{a}(|k+1\rangle\langle k+1| - |k\rangle\langle k|) \\ & + \sum_k \frac{(e_k - 2e_{k+1} + e_{k+2})g_{k,k+1}g_{k+1,k+2}}{(\omega_r + e_k - e_{k+1})(2\omega_r + e_k - e_{k+2})(\omega_r + e_{k+1} - e_{k+2})} \hat{a}^\dagger |k\rangle\langle k+2| \end{aligned} \quad (3.27)$$

and at $\mathcal{O}(\lambda^3)$

$$\hat{a}^{(3)} = \sum_k \left[\Lambda_k^{(1)} \hat{a}^2 |k+1\rangle\langle k| + \Lambda_k^{(2)} \hat{a}^\dagger \hat{a} |k\rangle\langle k+1| + \Lambda_k^{(3)} (\hat{a}^\dagger)^2 |k\rangle\langle k+3| + \Lambda_k^{(4)} |k\rangle\langle k+1| \right] \quad (3.28)$$

Now note for $\mathcal{O}(\lambda^2)$ we have a dephasing term in the first term and decay from $k+2 \rightarrow k$ mediated via a creation of a single photon and at higher order we observe processes mediated via two photons and last term is similar to what we observed at $\mathcal{O}(\lambda)$.

Note that the sum of above terms show what a transformed cavity annihilation operator looks like in the dispersive frame and on the very first look one can deduce the kind of process that occurs in the dissipator and one realises there are processes not only the cavity but also the transmon and sometimes these processes on transmon are mediated via photon processes in the cavity.

For the drive in the transformed frame, we'll have additional terms as hermitian conjugate of the terms in the above equation which look at $\mathcal{O}(\lambda)$ as a displacement operator on the qubit and cavity acting separately but at higher orders we have terms such as $\hat{a} |k+1\rangle\langle k| + \hat{a}^\dagger |k\rangle\langle k+1|$ which are similar to two-mode squeezing terms and higher orders we have terms which are much more complex.

3.3 Microwave Field Detection and Readout

In earlier section we discussed that for a transmon dispersively coupled to a readout cavity there is a frequency shift in the cavity dependent on the state of the transmon denoted as χ_k in Eq (3.22) which also include a higher order corrections. The readout circuit can be setup to measure either reflection or transmission of a signal. The best state discrimination is obtained by maximizing the separation between the two states in the I-Q plane. By fixing the cavity drive frequency we can map the qubit dynamics onto the phase of classical microwave response using homodyne detection.

3.3.1 I-Q Plane

In the earlier section we talked about the general principle on how to take a readout of superconducting qubit which includes separating the two states in the IQ plane. IQ plane is a two dimensional

way to represent the quantum information using quasi-probability distribution. For a quantum harmonic oscillator we define the field operators as $\hat{a} \sim \hat{x} + i\hat{p}$ where $\hat{x}$ and $\hat{p}$ are conjugate variables and follow Hisenberg's uncertainty relation $\Delta x \Delta p \geq \frac{1}{2}$. We can extend similar ideas to define quadrature operators as [Wall 08]

$$\hat{I} = \frac{i(\hat{a}^\dagger - \hat{a})}{2} \quad (3.29)$$

$$\hat{Q} = \frac{(\hat{a}^\dagger + \hat{a})}{2} \quad (3.30)$$

3.3.2 Field Detection

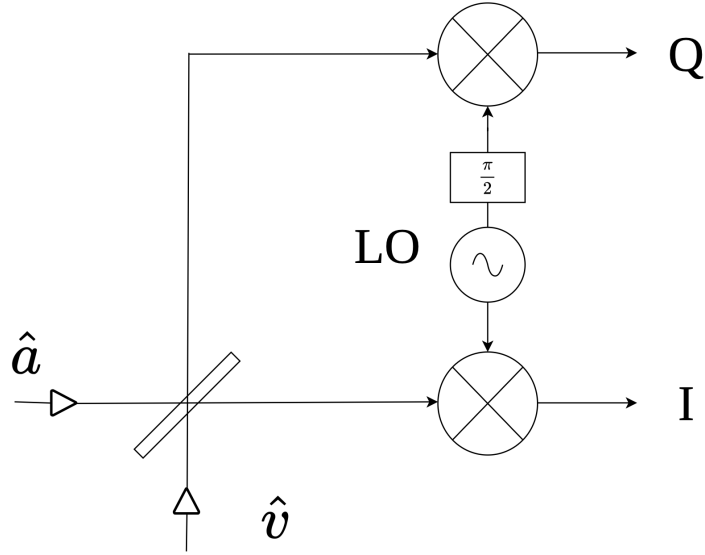


Figure 3.1: Schematic Representation of an IQ mixer. The rf signal from the readout setup is divided into two parts using the power divider shown by a beam splitter and it accounts for the added noise at this stage. The output signal is then mixed with a LO at mixers and by shifting the phase of the LO by $\frac{\pi}{2}$ in one of the arms we can simultaneously measure the two quadratures of the field.

Consider the output signal from the readout setup as a classical field given as $V_{rf} = A_{rf} \cos(\omega_{rf}t + \phi_{rf})$ mixed with the LO consist of a reference signal with well defined $V_{LO} = A_{LO} \cos(\omega_{LO}t - \phi_{LO})$, the output at mixers is given as

$$V_{mix} = \frac{V_{rf}V_{LO}}{2} [\cos((\omega_{LO} - \omega_{rf})t - \phi_{rf} - \phi_{LO}) + \cos((\omega_{rf} + \omega_{LO})t + \phi_{rf} + \phi_{LO})] \quad (3.31)$$

Now setting up $\phi_{rf} = 0$ using the convention in [Pozar 11], the output signal from the mixer is given as

$$V_{mix} = \frac{V_{rf}V_{LO}}{2} [\cos((\omega_{LO} - \omega_{rf})t - \phi_{LO}) + \cos((\omega_{rf} + \omega_{LO})t - \phi_{LO})] \quad (3.32)$$

Now the high frequency component is filtered out using a low-pass filter leaving the component $\omega_{IF} = (\omega_{LO} - \omega_{rf})$. The choice that $\omega_{IF} \neq 0$ corresponds to heterodyne detection and $\omega_{IF} = 0$ for homodyne detection. For $\phi_{LO} = 0$ it corresponds to signal in phase and $\phi_{LO} = \frac{\pi}{2}$ it corresponds to quadrature signal.

In quantum mechanical description of homodyne detection, leaving the ϕ_{LO} arbitrary the time-independent IF signal quadrature is proportional to

$$\hat{X}_{LO} = \frac{\hat{a}^\dagger e^{i\phi_{LO}} + \hat{a} e^{-i\phi_{LO}}}{2} = \hat{X} \cos \phi_{LO} + \hat{P} \sin \phi_{LO} \quad (3.33)$$

The homodyne signal, in principle is possible but it is susceptible to $\frac{1}{f}$ from electronics hence leading to lower SNR and it is not possible to do frequency division multiplexing(FDM)[[Krantz 19](#)].

3.4 Coherent States and Their Properties

Coherent states, universally denoted as $|\alpha\rangle$, are the quantum states that most closely approximate classical electromagnetic waves. They are fundamentally defined as the right eigenstates of the non-Hermitian annihilation operator $\hat{a}$:

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle, \quad \text{where } \alpha \in \mathbb{C}. \quad (3.34)$$

Because $\hat{a}$ is not a Hermitian operator, its eigenvalues α are continuous complex numbers. In polar form, $\alpha = |\alpha|e^{i\theta}$, where $|\alpha|$ relates to the amplitude of the optical field and θ is its phase.

3.4.1 Expansion in the Fock Basis

To express the coherent state in terms of discrete photon number (Fock) states $|n\rangle$, we apply the definition in Equation 3.34 alongside the standard action of the annihilation operator, $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$. Solving the resulting recursion relation yields a Poissonian superposition of number states:

$$|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \quad (3.35)$$

From this expansion, the probability $P(n)$ of measuring exactly n photons in a coherent state $|\alpha\rangle$ strictly follows a Poisson distribution:

$$P(n) = |\langle n|\alpha\rangle|^2 = \frac{|\alpha|^{2n} e^{-|\alpha|^2}}{n!}. \quad (3.36)$$

The mean photon number is simply the squared magnitude of the complex amplitude: $\langle \hat{n}|\hat{n}\rangle = \langle \hat{a}^\dagger \hat{a}|\hat{a}^\dagger \hat{a}\rangle = |\alpha|^2$.

3.4.2 Minimum Uncertainty

Coherent states are minimum-uncertainty wavepackets. If we define the dimensionless position and momentum quadrature operators as $\hat{X} = (\hat{a} + \hat{a}^\dagger)/\sqrt{2}$ and $\hat{P} = (\hat{a} - \hat{a}^\dagger)/(i\sqrt{2})$, the variances for a

coherent state are equal and minimize the Heisenberg uncertainty relation:

$$\Delta X^2 = \Delta P^2 = \frac{1}{2} \implies \Delta X \Delta P = \frac{1}{2}. \quad (3.37)$$

In phase space, this equal uncertainty manifests as a symmetric, circular Gaussian distribution centered at the classical coordinate $(\text{Re}(\alpha), \text{Im}(\alpha))$.

3.4.3 Non-Orthogonality and Overcompleteness

Unlike the eigenstates of Hermitian operators (such as the energy eigenstates $|n\rangle$), coherent states are not orthogonal to one another. The inner product of two distinct coherent states $|\alpha\rangle$ and $|\beta\rangle$ is:

$$\langle\beta|\alpha\rangle = \exp\left(-\frac{1}{2}|\alpha|^2 - \frac{1}{2}|\beta|^2 + \beta^*\alpha\right). \quad (3.38)$$

The squared magnitude of their overlap decays exponentially with the square of their distance in the complex plane:

$$|\langle\beta|\alpha\rangle|^2 = \exp(-|\alpha - \beta|^2). \quad (3.39)$$

Therefore, two coherent states are only approximately orthogonal when they are macroscopic and widely separated ($|\alpha - \beta| \gg 1$). Despite this non-orthogonality, coherent states form an *overcomplete* basis that spans the Hilbert space, satisfying the continuous closure relation:

$$\frac{1}{\pi} \int |\alpha\rangle \langle\alpha| d^2\alpha = \hat{\mathbb{I}}, \quad (3.40)$$

where $d^2\alpha = d(\text{Re } \alpha) d(\text{Im } \alpha)$ and $\hat{\mathbb{I}}$ is the identity operator.

3.4.4 The Displacement Operator

Coherent states can be formally defined as displaced vacuum states. This is achieved using the unitary displacement operator, denoted as $\hat{D}(\alpha)$:

$$\hat{D}(\alpha) = \exp(\alpha \hat{a}^\dagger - \alpha^* \hat{a}). \quad (3.41)$$

Applying this operator to the vacuum state $|0\rangle$ generates the corresponding coherent state:

$$|\alpha\rangle = \hat{D}(\alpha) |0\rangle. \quad (3.42)$$

The displacement operator possesses several fundamental mathematical properties that dictate how coherent states behave in phase space:

- **Unitarity and Inversion:** The displacement operator is unitary. Its Hermitian adjoint is equivalent to a displacement in the exact opposite direction in phase space:

$$\hat{D}^\dagger(\alpha) = \hat{D}^{-1}(\alpha) = \hat{D}(-\alpha). \quad (3.43)$$

- **Operator Transformation:** Applying the displacement operator to the annihilation and creation operators shifts them by the complex amplitude α . Using the Baker-Campbell-Hausdorff lemma, the transformation yields:

$$\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha, \quad (3.44)$$

$$\hat{D}^\dagger(\alpha)\hat{a}^\dagger\hat{D}(\alpha) = \hat{a}^\dagger + \alpha^*. \quad (3.45)$$

This property highlights why $|\alpha\rangle$ is called a displaced state—the operator literally translates the electromagnetic field operators by a classical, macroscopic amount α .

- **Composition Rule:** Two successive displacements do not simply add their amplitudes because $\hat{a}$ and $\hat{a}^\dagger$ do not commute (their commutator is $[\hat{a}, \hat{a}^\dagger] = 1$). Multiplying two displacement operators introduces a geometric phase factor:

$$\hat{D}(\alpha)\hat{D}(\beta) = \exp\left[\frac{1}{2}(\alpha\beta^* - \alpha^*\beta)\right]\hat{D}(\alpha + \beta). \quad (3.46)$$

Because the argument of the leading exponential $\frac{1}{2}(\alpha\beta^* - \alpha^*\beta)$ is purely imaginary, this extra term represents an overall phase shift rather than a change in the physical amplitude of the state.

3.4.5 Phase-Space Representation and Field Detection

In the context of field detection, it is highly useful to employ quasiprobability distributions such as the Wigner function and the Husimi Q-function to visualize quantum states in a classical-like phase space. In this section, we define these distributions and demonstrate their form for a fundamental optical state.

The Wigner function is formally defined as the two-dimensional Fourier transform of the symmetrically ordered characteristic function. The characteristic function is given by:

$$\chi_S(\eta) = \langle e^{\eta\hat{a}^\dagger - \eta^*\hat{a}} \rangle = \text{Tr}\left\{\rho e^{\eta\hat{a}^\dagger - \eta^*\hat{a}}\right\}. \quad (3.47)$$

From this, the Wigner function is constructed as:

$$W(\alpha) = \frac{1}{\pi^2} \int d^2\eta \chi_S(\eta) e^{\eta^*\alpha - \eta\alpha^*}. \quad (3.48)$$

While the Wigner function successfully yields the correct quantum marginal distributions when integrated over a quadrature, it is not strictly positive definite for all quantum states. Consequently, it is a quasiprobability rather than a true probability distribution. However, unlike the Glauber-Sudarshan P-representation, the Wigner function is bounded and finite at every point in phase space [Wall 08].

The Husimi Q-function offers an alternative representation, defined simply as the expectation value of the density operator ρ evaluated in the coherent state basis:

$$Q(\alpha) = \frac{1}{\pi} \langle \alpha | \rho | \alpha \rangle. \quad (3.49)$$

Due to the completeness relation of coherent states, $\frac{1}{\pi} \int d^2\alpha |\alpha\rangle\langle\alpha| = \mathbb{I}$, and the trace condition $\text{Tr}(\rho) = 1$, the Q-function is normalized to unity. Furthermore, because it is an expectation value of a positive-semidefinite operator, $Q(\alpha)$ is strictly non-negative and bounded [Scully 97], making it a true probability distribution related to heterodyne measurement.

In context to dispersive qubit measurement, the Wigner function is particularly useful because it is related to the probability distribution of measurement of $\hat{X}$ and $\hat{P}$ quadrature. The marginals are defined as

$$P(x) = \int_{-\infty}^{\infty} dp W(x, p) = \langle x | \rho | x \rangle \quad (3.50)$$

$$P(p) = \int_{-\infty}^{\infty} dx W(x, p) = \langle p | \rho | p \rangle \quad (3.51)$$

This implies that marginal in one quadrature can be achieved by integrating over the other quadrature.

Example: Distributions for a Coherent State

As a simple example, we illustrate the case of a pure coherent state $|\alpha_0\rangle$. In this scenario, the state perfectly mimics classical behavior, and both distributions take the form of simple, isotropic Gaussians.

Applying the definition in Eq (3.48) to the state $\rho = |\alpha_0\rangle\langle\alpha_0|$, the resulting Wigner function reveals the symmetric, minimum-uncertainty noise footprint of the state:

$$W(\alpha) = \frac{2}{\pi} \exp(-2|\alpha - \alpha_0|^2). \quad (3.52)$$

Because the state is entirely classical in nature, its Wigner function is positive everywhere ($W \geq 0$).

Conversely, using the definition in Eq (3.49), the Q-function for the same coherent state $|\alpha_0\rangle$ evaluates directly to the modulus squared of the coherent state overlap:

$$Q(\alpha) = \frac{1}{\pi} |\langle \alpha | \alpha_0 \rangle|^2 = \frac{1}{\pi} \exp(-|\alpha - \alpha_0|^2). \quad (3.53)$$

Comparing Eq (3.52) and Eq (3.53), we see that both distributions are centered at the classical amplitude α_0 . However, the Q-function is noticeably broader (its exponential argument is smaller by a factor of 2). This additional broadening reflects the penalty of vacuum noise introduced when simultaneously attempting to measure both conjugate quadratures of the field.

Since both $Q(\alpha)$ and $W(\alpha)$ give the complete description of the quantum state it is possible to connect the two. For example a Q function can be written as a convolution of Wigner function with a gaussian

$$Q(\alpha) = \frac{2}{\pi} \int_{-\infty}^{\infty} d^2\beta W(\beta) e^{-2|\alpha - \beta|^2} \quad (3.54)$$

In other words $Q(\alpha)$ is obtained by adding noise to the Wigner function which explains the broader variance for Eq (3.53) and hence the measurement outcome after the homodyne detection are much better given by marginals of Q function than of the Wigner function due to added noise at power divider as shown in Figure 3.1.

3.5 Optimal Condition for the Dispersive Readout

The objective of a qubit readout is to maximize the readout fidelity in the shortest measurement time possible [Blais 21]. To achieve this goal we'll observe the evolution of the intracavity field using Langevin Equation [Wall 08]. Consider the dispersive hamiltonian in Eq (2.27) with an additional drive term as $\Omega(t)(\hat{a}^\dagger + \hat{a})$, using Hisenberg's equation of motion for intracavity field $\hat{a}$,

$$\dot{\hat{a}} = -i(\delta_r + \chi\sigma_z)\hat{a} - i\Omega(t) - \frac{\kappa}{2}\hat{a} \quad (3.55)$$

where $\delta_r = \omega_r - \omega_d$ Now taking expectation value of $\hat{a}$ and dropping the time dependence of the drive $\Omega(t) \rightarrow \Omega$, the equation of motion for the complex amplitude $\alpha_{0,1}$ follow a langevin equation given as

$$\dot{\alpha}_{0,1} = -i(\delta_r \pm \chi)\alpha_{0,1} - i\Omega(t) - \frac{\kappa}{2}\alpha_{0,1} \quad (3.56)$$

and the steady state solution for the above equation is

$$\alpha_{0,1} = \frac{-\Omega}{(\delta \pm \chi) - i\frac{\kappa}{2}} \quad (3.57)$$

Now the $\text{Re}\{\alpha_{0,1}\} = \frac{-\Omega(\delta \pm \chi)}{(\delta \pm \chi)^2 + (\frac{\kappa}{2})^2}$ and $\text{Im}\{\alpha_{0,1}\} = \frac{\Omega(\frac{\kappa}{2})}{(\delta \pm \chi)^2 + (\frac{\kappa}{2})^2}$ now setting the $\delta = 0$ one realise that all the information about the state is found only along X quadrature. Here we can also define the steady state phase amplitude and phase as

$$A_{0,1}^s = \sqrt{\langle \hat{X}_{0,1} \rangle^2 + \langle \hat{P}_{0,1} \rangle^2} = \frac{\Omega}{\sqrt{(\kappa/2)^2 + (\delta \pm \chi)^2}} \quad (3.58)$$

and the phase is

$$\phi_{0,1}^s = \arctan\left(\frac{\langle \hat{X}_{0,1} \rangle}{\langle \hat{P}_{0,1} \rangle}\right) = \arctan\left(\frac{\delta \pm \chi}{\kappa/2}\right) \quad (3.59)$$

At this point we realise that for coherent tone of frequency of $\omega_r \pm \chi$ will be reflected or transmitted depending upon the state of the qubit. Alternatively, we can also drive the qubit at the bare cavity frequency (ω_r) and there is a phase accumulation in the transmitted signal depending upon the state of the qubit and the phase shift for $\delta = 0$ is simply $\pm \arctan\left(\frac{2\chi}{\kappa}\right)$. As a result, in the dispersive regime by measuring the amplitude or the phase of the transmitted signal from the resonator reveals the information about the qubit state [Blais 04][Blais 21].

Now we define Signal-to-Noise Ratio (SNR) as a metric for a good readout given as

$$\text{SNR}(t) = \frac{|\langle \hat{M}_1 \rangle - \langle \hat{M}_0 \rangle|}{\sqrt{\langle \hat{M}_{N0} \rangle^2 + \langle \hat{M}_{N1} \rangle^2}} \quad (3.60)$$

where $\hat{M}(\tau) = \sqrt{\kappa} \int_0^\tau dt [\hat{a}_{out}^\dagger(t)e^{i\phi_h} + \hat{a}_{out}(t)e^{-i\phi_h}]$ is the measurement operator and ϕ_h is the homodyne angle. The signal for a measurement is given as $\langle \hat{M} \rangle$ while for simplicity we assume $\langle \hat{M}_N^2 \rangle = \kappa\tau$

The langevin equation for the cavity field in the interaction picture reads as

$$\dot{\hat{a}} = -i\chi\sigma_z\hat{a} - \frac{\kappa}{2}\hat{a} - \sqrt{\kappa}\hat{a}_{in} \quad (3.61)$$

where $\hat{a}_{in}$ is the input field and the field is defined using its mean as $\langle \hat{a}_{in} \rangle = \alpha_{in} = -\frac{\Omega}{\sqrt{\kappa}}$ and $\Omega = |\Omega|e^{i\phi_d}$ and integrating the langevin equation and using input output relation $\hat{a}_{out} - \hat{a}_{in} = \sqrt{\kappa}\hat{a}$ [Yurke 04]

$$\alpha_{out} = \frac{\Omega}{\sqrt{\kappa}} e^{-i\phi_{qb}\langle\sigma_z\rangle} \left[1 - 2e^{-(i\chi\langle\sigma_z\rangle + \frac{1}{2}\kappa)t + \frac{1}{2}i\phi_{qb}\langle\sigma_z\rangle} \right] \quad (3.62)$$

where $\phi_{qb} = \frac{2\chi}{\kappa}$ Using the above expression to compute numerator in the SNR as

$$M_{|1\rangle} - M_{|0\rangle} = 4|\Omega| \sin\phi_{qb} \sin(\phi_d - \phi_h) \tau \left\{ 1 - \frac{4}{\kappa\tau} \left[\cos^2 \frac{1}{2}\phi_{qb} - \frac{\sin(\chi\tau + \phi_{qb})}{\sin(\phi_{qb})} e^{-\frac{1}{2}\kappa\tau} \right] \right\} \quad (3.63)$$

The measurement signal is optimized for $\phi_d - \phi_h = \frac{\pi}{2} \text{ mod } \pi$ and at long integration time $\phi_{qb} = \frac{\pi}{2}$ or equivalently [Didier 15]

$$\chi = \frac{\kappa}{2} \quad (3.64)$$

such an optimal condition can also be achieved for measurement time and along with the above result can be derived using quantum trajectories approach [Gambetta 08].

Now in the above discussion we restricted ourselves to a two level system and hence found that the state dependent dispersive shift is equal and opposite. For a multilevel anaharmonic system such as transmon, the state dependent shifts are not equal and have the same sign hence affecting the state discrimination using phase of the output signal. But to arrive at equal and opposite sign for two states it is possible by driving the cavity at different frequency other than ω_r and to get the drive frequency (ω_d) consider Eq (3.56) but $\pm\chi \rightarrow \chi_{0,1}$ and hence the equation is

$$\dot{\alpha}_{0,1} = -i(\delta_r + \chi_{0,1}) \alpha_{0,1} - i\Omega(t) - \frac{\kappa}{2} \alpha_{0,1} \quad (3.65)$$

and we want $\delta_r + \chi_0 = -(\delta_r + \chi_1)$ and hence $\delta_r = -\frac{\chi_0 + \chi_1}{2}$ and hence $\omega_d = \omega_r + \frac{\chi_0 + \chi_1}{2}$, hence effective dispersive shift is $\tilde{\chi} = \frac{\chi_0 - \chi_1}{2}$ which modify the optimal condition in Eq (3.64) as $\kappa = 2\tilde{\chi}$.

Chapter 4

Errors During the Readout

One of the main challenges to build a scalable fault-tolerant quantum computer is the ability to speed up measurement and initialization [Girvin 23]. In the current state of art it is possible to apply single or two qubit gates as fast as $\sim 10ns$ but measurement is still an order of magnitude slower. Another critical complication with the readout is residual photons left in the cavity and then causing the change in the qubit frequency due to ac-stark shift as shown in Sections 3.2 and 1.2.2 which leads to holding up the operations for $\tau_c \sim \frac{1}{\kappa}$, where κ is the cavity decay rate. Another source of error during the readout is the qubit driving as shown in Section 3.2.1 leading to a residual population in the different transmon levels. Most important thing during the readout is the ability to determine the state determined using readout fidelity and Signal-to-noise Ratio (SNR). In this chapter we'll explore all these errors that can occur during the readout. We'll also discuss a readout technique presented in [Jerger 24] meant for residual cavity photon annulment and its impact on qubit error and SNR.

4.1 Readout with Cavity Reset

In the dispersive limit the dynamics of a transmon coupled to a readout cavity can be well approximated to second order using hamiltonian given as

$$\hat{H} = \omega_r \hat{a}^\dagger \hat{a} + e_k |k\rangle\langle k| + \chi_k \hat{a}^\dagger \hat{a} |k\rangle\langle k| \quad (4.1)$$

which in interaction picture is given as

$$\hat{H}_I = \chi_k \hat{a}^\dagger \hat{a} |k\rangle\langle k| \quad (4.2)$$

now using langevin equation with an input field $\hat{a}_{in}$ as

$$\dot{\hat{a}} = -i\chi \hat{a} |k\rangle\langle k| - \frac{\kappa}{2} \hat{a} + \sqrt{\kappa} \hat{a}_{in} \quad (4.3)$$

in the presence of a coherent drive all the operator expectation values can be treated as a complex amplitude

$$\dot{a} = -i\chi_k a - \frac{\kappa}{2} + \sqrt{\kappa} a_{in} \quad (4.4)$$

now defining the Fourier transform as $a(t) = \int a(\omega) e^{-i\omega t} d\omega$ and taking fourier transform of the above equation we have

$$-i\omega a(\omega) + i\chi_k a(\omega) + \frac{\kappa}{2} a(\omega) = \sqrt{\kappa} a_{in}(\omega) \quad (4.5)$$

hence we can write the reponse of the readout cavity to the input field a_{in} as

$$a(\omega) = \frac{\sqrt{\kappa}}{\frac{\kappa}{2} - i(\omega - \chi_k)} a_{in}(\omega) = h(\omega) a_{in}(\omega) \quad (4.6)$$

building on the above discussion it is possible to tailor the input signal such that we achieve the reset of the readout cavity. To achieve a desired reponse we choose a trial function, $a = a_T(t)$ such that $a_T = 0 \forall t \leq 0, t \geq T_p$. The input field that generates such reponse for the ground state is given as

$$a_{in}(\omega) = h^{-1}(\omega) a_T(\omega) \quad (4.7)$$

By taking the inverse fourier transform

$$a_{in}(t) = \frac{1}{\sqrt{\kappa}} \left[\frac{\kappa}{2} + i\chi_0 + \frac{d}{dt} \right] a_T(t) \quad (4.8)$$

This technique is known as Derivative Removal for Annulment of Cavity-Hybridized Measurement Adiabats (drachma). The above technique can be generalised to N states in the following way

$$a_{in}(t) = \frac{1}{\sqrt{\kappa^N}} \prod_{k=0}^{N-1} \left[\frac{\kappa}{2} + i\chi_k + \frac{d}{dt} \right] a_T(t) \quad (4.9)$$

A similar analysis is also present when there are strong nonlinearities present in the system [Jerger 24].

4.2 DRACHMA in Action

In the earlier section we presented the technique which in principle should address the problem of residual photons in the cavity and in this section we'll check the efficiency of the technique in different coupling regimes. As defined in Section 2.3 we defined n_{crit} which in principle tells the maximum number of photons present in the cavity beyond which the dispersive approximation breaks. Given, $n_{crit} = \left(\frac{\Delta}{2g}\right)^2$ for larger g we have smaller n_{crit} and also for larger detuning we have larger n_{crit} hence placing the cavity closer to the qubit means more dressing between adjacent level connected by single photon process as shown in Eqs (1.18 & 1.19). Hence smaller n_{crit} means larger coupling or cavity place closely to qubit frequency and in this work we'll associate n_{crit} with qubit cavity coupling.

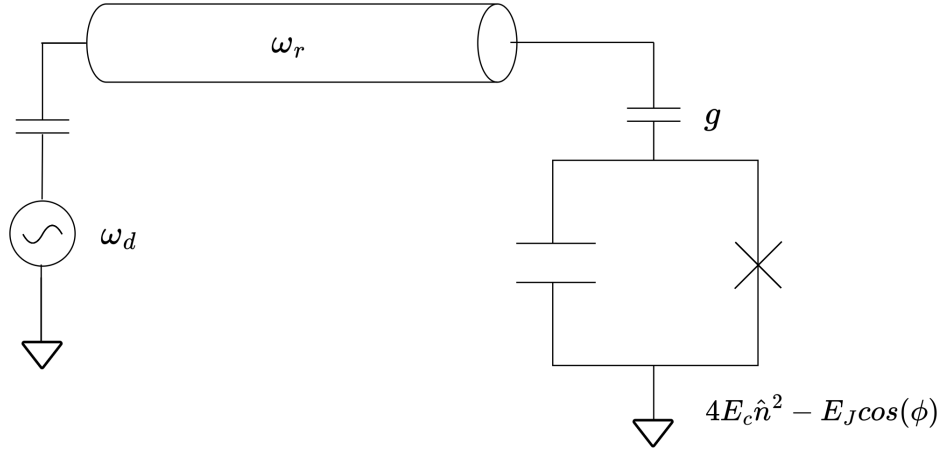


Figure 4.1: Circuit representation of readout circuit

4.2.1 System parameters

For the purpose of this we'll simulate the circuit in Figure 4.1 where the transmon parameters are given in Table 4.1

Circuit Parameter	Value (GHz)
Charging Energy ($E_C/2\pi$)	0.166
Josephson Energy ($E_J/2\pi$)	20.057
Qubit Frequency ($\omega_q/2\pi$)	5.005
Anharmonicity ($\alpha/2\pi$)	-0.181

Table 4.1: Transmon specifications used in the simulation

While dispersively coupled to the readout cavity via coupling strength $g_{k,k+1}$ with some detuning with the cavity given as $\Delta = \omega_r - \omega_q$ we can achieve different n_{crits} by fixing one of detuning or coupling and varying the another or by varying both of them to get a desired ratio. In our work we'll fix the coupling and vary the qubit-cavity detuning as it seems more practical from an experimental point of view. In Table 4.2 we have listed the value of important energy shifts and rates for a dispersively coupled transmon cavity for different n_{crits}

4.2.2 Photon Number Dynamics in the Cavity

In the earlier sections we talked about the challenge of residual photons in the cavity and engineering the pulse so that we can actively remove the photons from the cavity. For the

Parameter	$n_{crit} = 5$ (GHz)	$n_{crit} = 10$ (GHz)	$n_{crit} = 15$ (GHz)
Cavity Frequency ($\omega_r/2\pi$)	6.313	6.863	7.280
Cavity Qubit detuning ($\Delta/2\pi$)	1.307	1.857	2.75
Ground State Dispersive Shift ($\chi_0/2\pi$)	0.065	0.046	0.037
Excited State Dispersive Shift ($\chi_1/2\pi$)	0.045	0.035	0.029
Cavity Driving Frequency ($\omega_d/2\pi$)	6.369	6.904	7.314
Effective Dispersive Shift ($\tilde{\chi}/2\pi$)	0.01	0.05	0.025
Cavity Decay Rate ($\kappa/2\pi$)	0.02	0.1	0.051
Critical Amplitude for Flat-top ($\Omega_c^{(F)}/2\pi$)	0.029	0.023	0.021
Critical Amplitude for DRACHMA ($\Omega_c^{(D)}/2\pi$)	0.332	0.3	0.278

Table 4.2: Critical parameters for dynamics of dispersively coupled transmon-cavity interaction

purpose of this work we'll use a Flat-top pulse given as

$$\Omega(t) = \Omega_0 \text{erf}\left(\frac{t}{t_r}\right)^4 \text{erf}\left(\frac{T_m - t}{t_r}\right)^4 \quad (4.10)$$

where t_r is rasing time and T_m is the measurement time and for our work they are $10ns$ and $100ns$ respectively. The pulse shape for Flat-top and DRACHMA are shown in Figure 4.2 and Figure 4.3 respectively for $n_{crit} = 5$ parameters given in Table 4.2.

In Figure 4.2 and Figure 4.3 one note that the signal goes to zero at the $100ns$ for both but from Figure 4.4 we realise that there is a lag between signal and photon population in the cavity for Flat-top pulse but with DRACHMA we ensure that there is reset of cavity by making the photon population in the cavity to zero.

In Figure 4.5 we have the residual photon population in the cavity for system starting in ground state and in the first excited state respectively and it important to note that number of residual photons for system starting in ground state are less than that for system starting in the first excited state for small pulse amplitudes but the trend change as we to higher pulse amplitudes and it happnes around $\Omega/\Omega_c = 0.3$ for $n_{crit} = 5$, $\Omega/\Omega_c = 0.2$ for $n_{crit} = 10, 15$. Something similar is also observed for DRACHMA in $n_{crit} = 5$ but the transition happens at much larger power, around $\Omega/\Omega_c = 0.9$, for $n_{crit} = 10$ we have more photons in the first excited state than ground state for all amplitudes but the numbers are fairly close to each other but for $n_{crit} = 15$ the gap grows and photon population in first excited state still more than the ground state and something similar in observed for larger n_{crit} in [Jerger 24].

We also explore a better metric for performance of DRACHMA as $\langle N_{res} \rangle / \langle N_{max} \rangle$ in

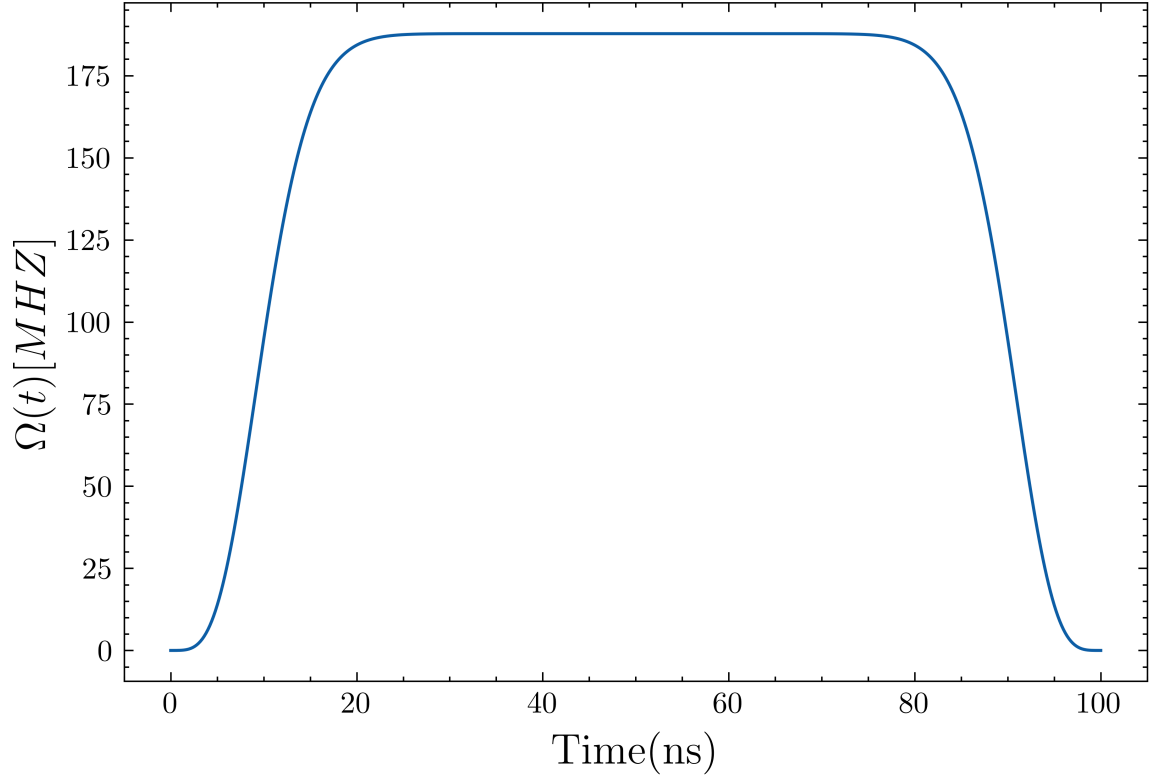


Figure 4.2: Flat-top pulse for $n_{crit} = 5$ parameters

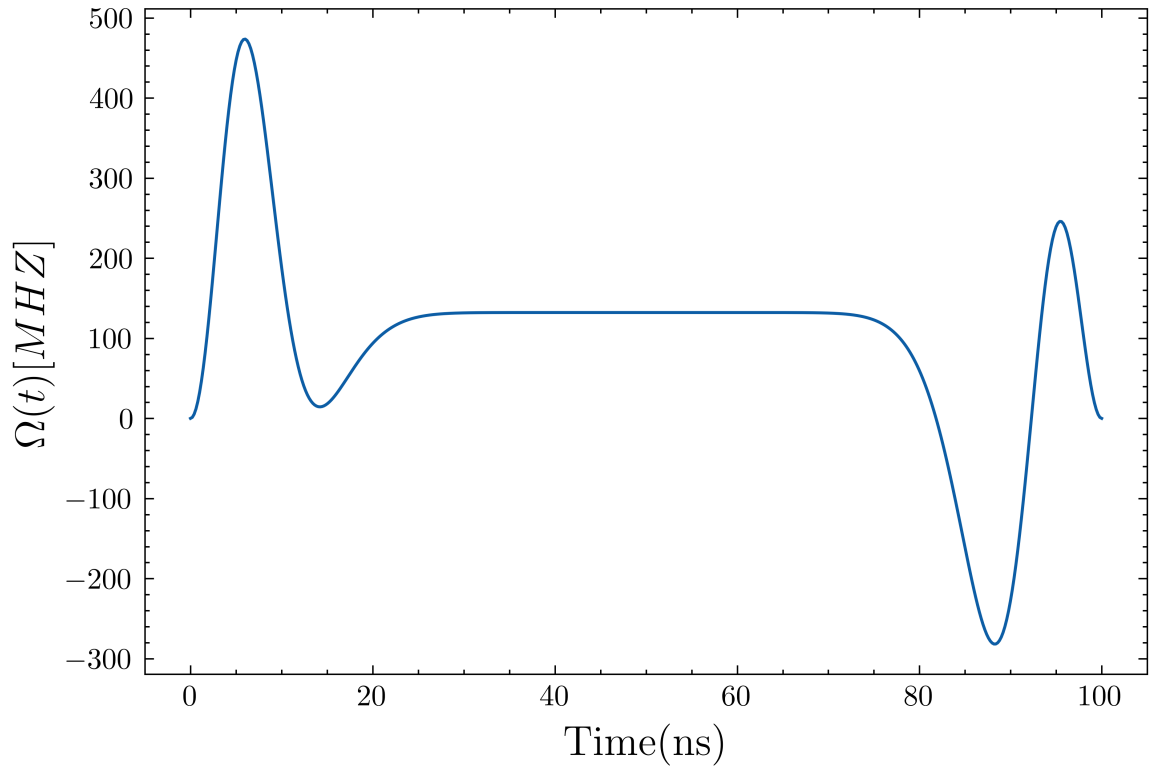
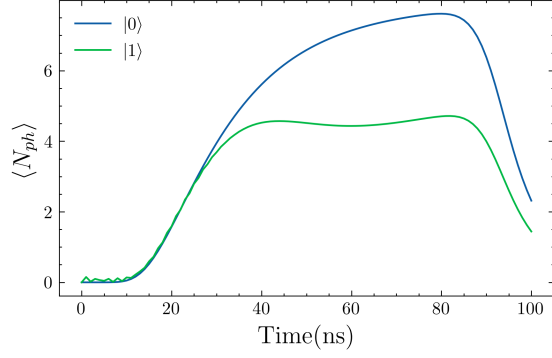
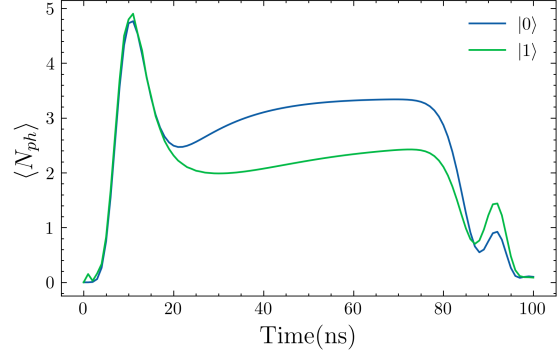


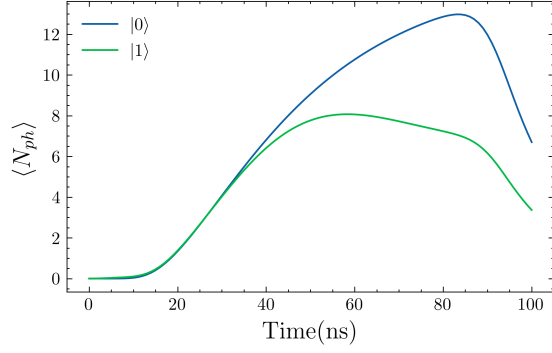
Figure 4.3: DRACHMA pulse for $n_{crit} = 5$ parameters



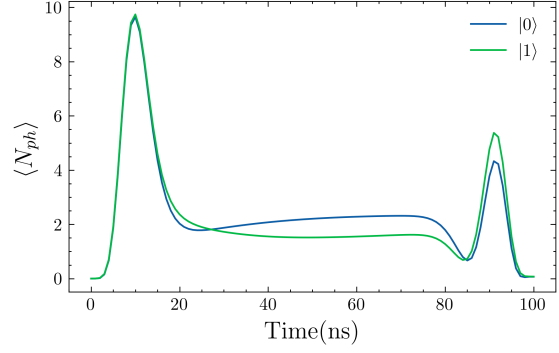
(a) $n_{crit} = 5$ Flat-top Pulse



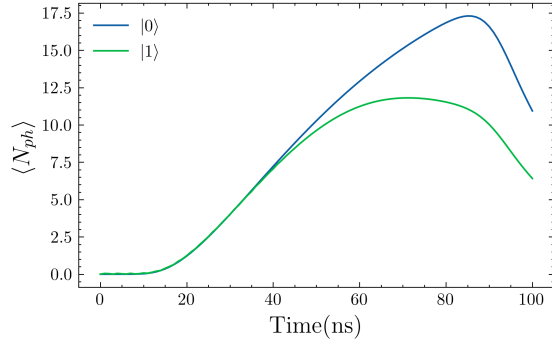
(b) $n_{crit} = 5$ DRACHMA Pulse



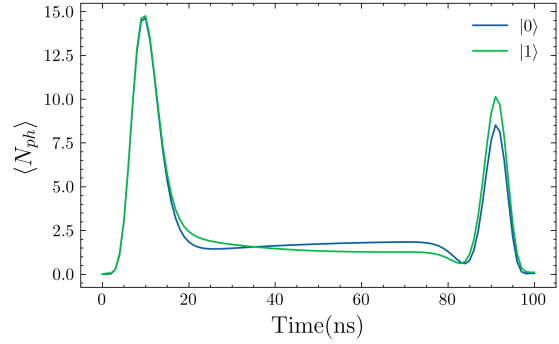
(c) $n_{crit} = 10$ Flat-top Pulse



(d) $n_{crit} = 10$ DRACHMA Pulse



(e) $n_{crit} = 15$ Flat-top Pulse



(f) $n_{crit} = 15$ DRACHMA Pulse

Figure 4.4: Photon dynamics in the cavity during the readout for different n_{crit} using DRACHMA and Flat-top for $\Omega = \Omega_c$

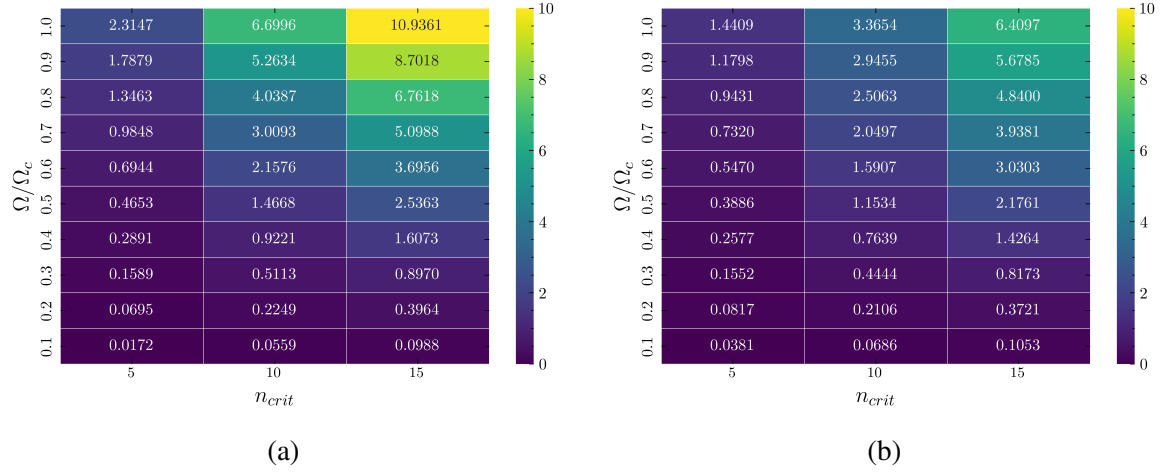


Figure 4.5: Residual Photons in the cavity with Flat-top pulse for system starting in (a) Ground State (b) First Excited State

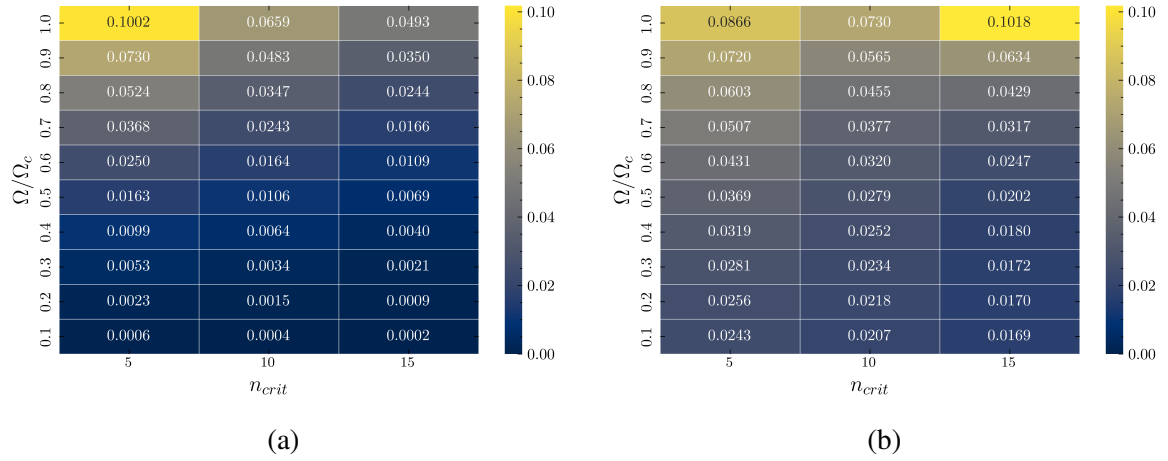


Figure 4.6: Residual Photons in the cavity with DRACHMA for system starting in (a) Ground State (b) First Excited State

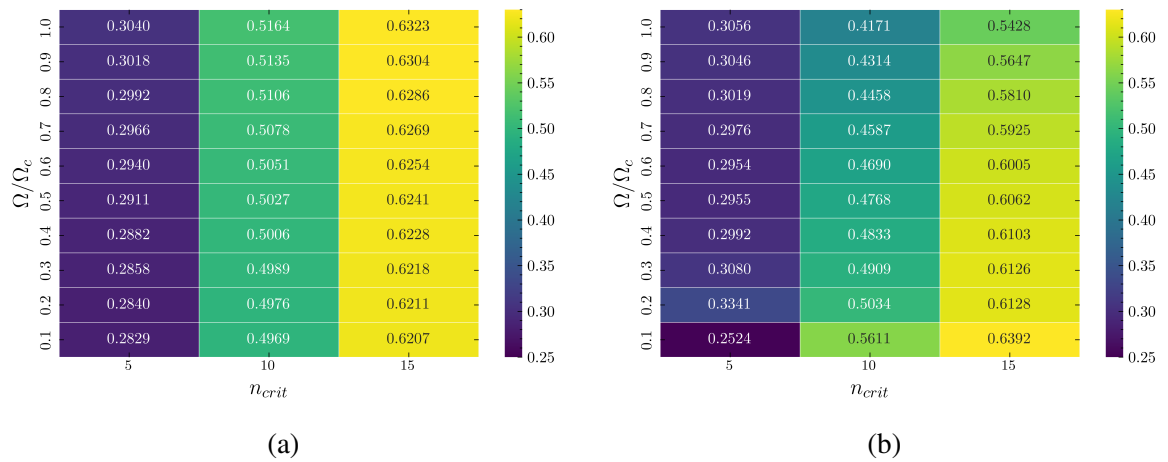


Figure 4.7: Ratio of Residual Photons to maximum number of photons in the cavity with Flat-top for system starting in (a) Ground State (b) First Excited State

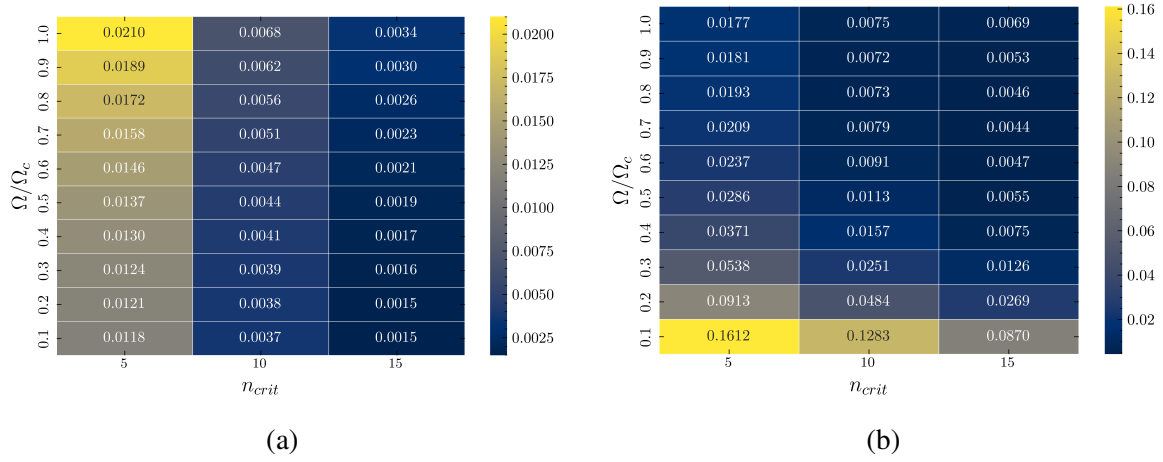


Figure 4.8: Ratio of Residual Photons to maximum number of photons in the cavity with DRACHMA for system starting in (a) Ground State (b) First Excited State

Figure 4.7 and Figure 4.8 and comparing the two we can conclude that DRACHMA performs much better than Flat-top and DRACHMA performs better as we move across n_{crit} as the system becomes more dispersive. For Flat-top, there is clear trend of increase in the ratio with pulse amplitude across the n_{crit} s but for the first excited state the trend is similar to ground state. For $n_{crit} = 5$, the trend shifts as we have move to $n_{crit} \in \{10, 15\}$ the trend reverse and the ratio goes from increasing across pulse amplitudes to decreasing. Similarly for DRACHMA, in the ground state, the ratio increase with pulse power across n_{crit} s but for the first excited state, the ratio decrease with the pulse power across n_{crit} s.

Nonetheless the DRACHMA has shown order of magnitudes improvement over Flat-top as shown in Figure 4.8 and Figure 4.7 and from Figure 4.6 and Figure 4.5 we observe photon populations for DRACHMA is roughly order of magnitude less than Flat-top which serves its purpose to reducing the waiting between measurement and further operations.

4.3 Qubit Error

As shown in Section 3.2.1 that the transmon also decays through the cavity and cavity driving also drives the qubit, leading to majorly two types of errors on the qubit as dissipative errors or incoherent errors and driving errors or coherent errors. To investigate the source of the error in more detail we'll work with pseudo frames such that first, we'll only transform the drive with the hamiltonian and not the dissipator and second, we'll transform the dissipator not the drive. The first one captures the effects of drive only on the qubit and second primarily captures the effect of the dissipator on the qubit. To capture the effects on the qubit we define a quantity the qubit error as

$$E_q = \frac{1}{2} (P(0|0) + P(1|1)) \quad (4.11)$$

Coupling ($g_{k,k+1}/2\pi$)	Value (GHZ)
$g_{0,1}$	0.293
$g_{1,2}$	0.407
$g_{2,3}$	0.489

Table 4.3: Coupling strength between $|k\rangle \rightarrow |k+1\rangle$

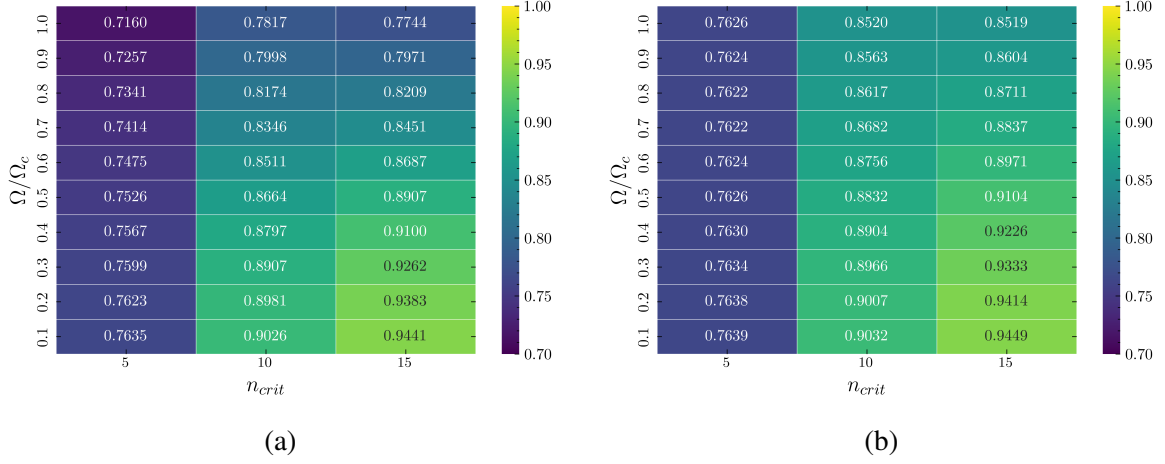


Figure 4.9: Qubit error in the labframe (a) Flat-top (b) DRACHMA

In Figures 4.9-4.11, we plot the qubit error as defined in Eq. (4.11) with n_{crit} and pulse amplitude up to Ω_c for different frames. It can be seen that at small amplitudes for both DRACHMA and Flat-top, the major source of error is dissipative in nature, as also shown in Figure 4.9 and Figure 4.11 where for $n_{crit} = 5$ the quantity remains fairly the same for DRACHMA and Flat-top as compared to other n_{crit} values. This is because at smaller n_{crit} , the rate of Purcell decay, $\gamma = \kappa \left(\frac{g}{\Delta}\right)^2$, is larger than for other n_{crit} values, as the factor follows:

$$\left(\frac{g}{\Delta}\right)^2 = \frac{1}{4n_{crit}} \quad (4.12)$$

hence the rate decrease as we move across the n_{crit} and the major source of error becomes drive dominated as we move up the pulse amplitude and across the n_{crit} as it can be seen in Figure 4.9 and Figure 4.10.

Another important conclusion that can be made from Figure 4.9 based on the difference in the qubit error for Flat-top and DRACHMA, one can conclude that DRACHMA also corrects the qubit errors and we can extend this to Figure 4.10 and Figure 4.11 where we realise that at larger amplitudes DRACHMA is much closer to 1, indicating that it also corrects the coherent errors and leaving the incoherent error which are primarily dissipative

in nature.

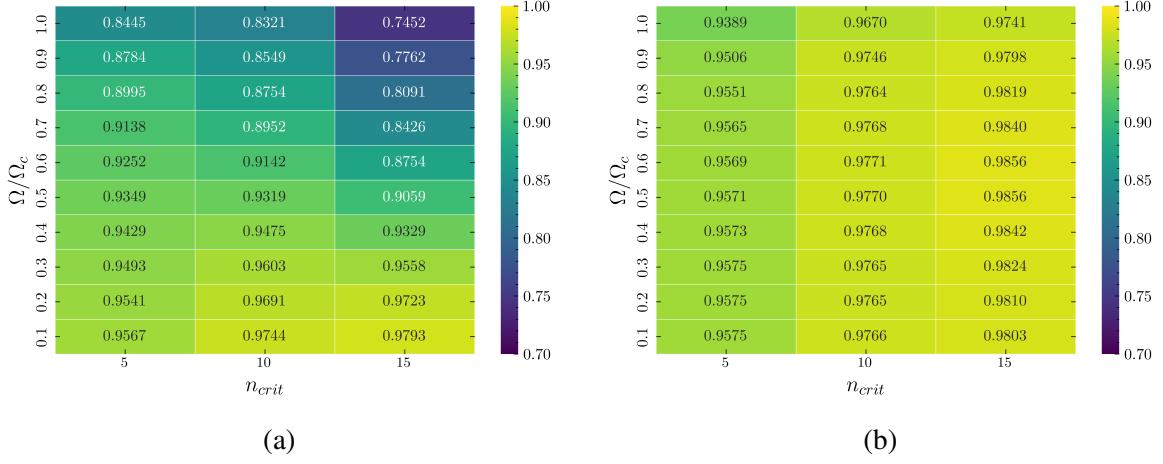


Figure 4.10: Qubit error due to driving (a) Flat-top (b) DRACHMA

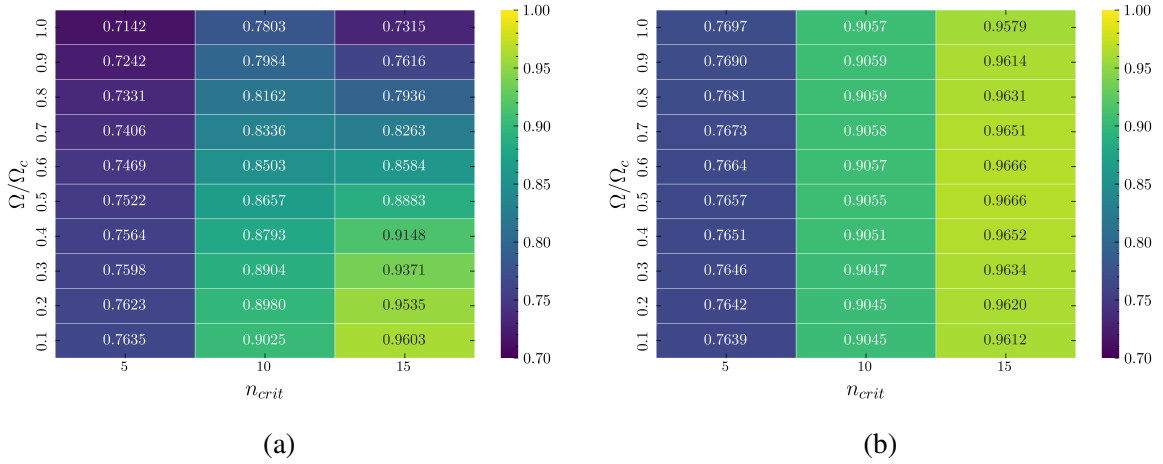


Figure 4.11: Qubit error due to dissipation (a) Flat-top (b) DRACHMA

4.4 Signal-to-Noise Ratio (SNR)

As defined in Eq (3.60), SNR serves as the measure using which one can compute the fidelity of the readout, where fidelity of the readout is given as

$$F = e^{-t/T_1} \text{erf} \left(\frac{SNR}{\sqrt{2}} \right) \quad (4.13)$$

where T_1 is the qubit lifetime and t is the signal integration time which for current parameters is $80ns$. In Figure 4.12 and Figure 4.13 we compare the Signal-to-Noise ratio for two pulses and SNRs are much better for DRACHMA are much better than the Flat-top as using DRACHMA the two pointer states corresponding to ground state and excited state move

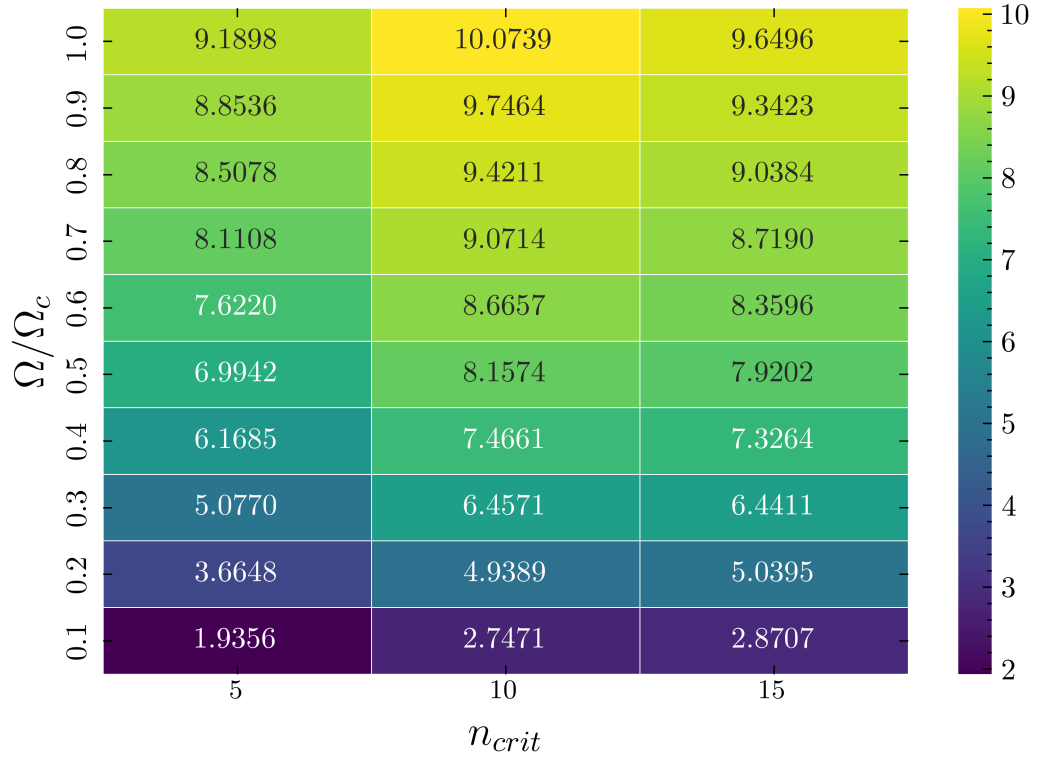


Figure 4.12: Signal-to-noise Ratio for Flat-top pulse

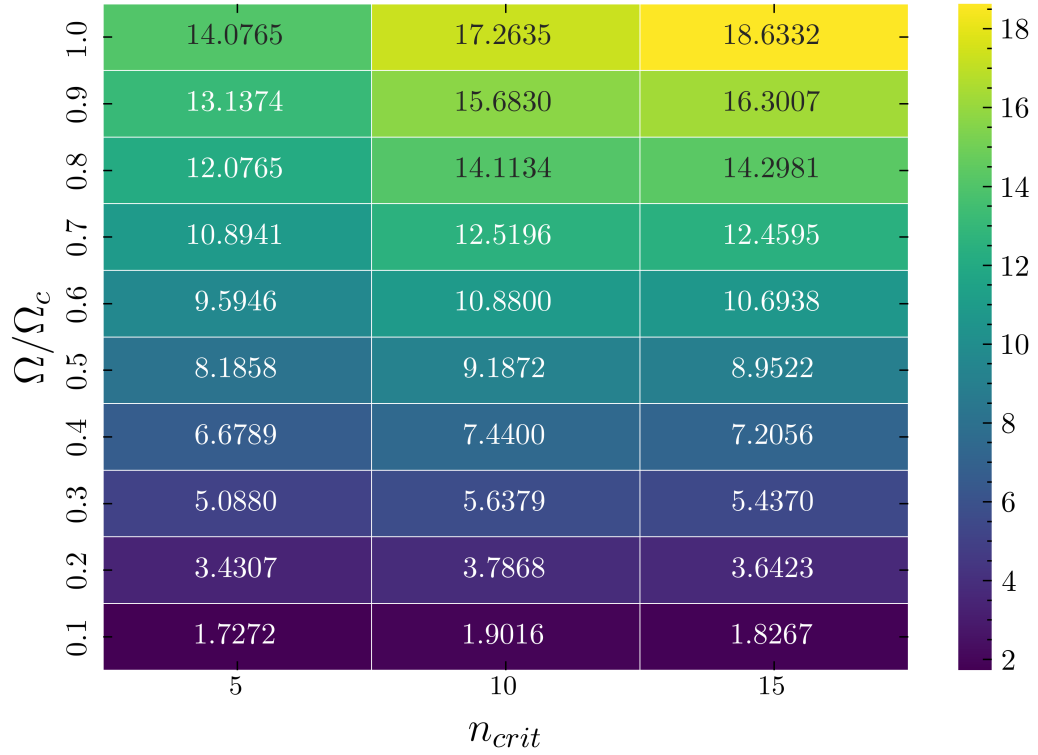


Figure 4.13: Signal-to-noise Ratio for DRACHMA

along one quadrature but in opposite directions hence SNRs for DRACHMA are much better than the Flat-top.

Chapter 5

Summary

In this work we investigated the driven-dissipative dynamics of the dispersively coupled transmon-cavity system. We employed both analytical methods such as Schrieffer Wolf Transformation, as well as numerical methods to investigate the system using lindblad master equation. n_{crit} is an important quantity in context of dispersive readout as it sets the convergence radius of the perturbation series. In work we have employed the n_{crit} as measure of coupling between the transmon and the readout cavity hence different n_{crits} show different kind of dynamics. One can track different quantities for the readout but in this we focused primarily on residual photon population, transmon population at the end of the readout and the signal-to-noise ratio. All three of them are important for a good readout the first two are important from the point of view of what we call as QNDness or Quantum Non Demolition behavior of the readout which allows for the cavity and qubit to return back to the initial state.

In this work used a Flat-Top pulse and pulse engineered using technique called as DRACHMA as compared the above three quantities for both of them across $n_{crit} \in \{5, 10, 15\}$ and pulse amplitude upto Ω_c . We showed in our work that DRACHMA performs better than flat-top pulse on all three metrics. We also observed that at larger amplitudes and at larger n_{crits} we have larger SNRs but this also leads to more qubit error and residual photons in the cavity hence creating a trade-off between the two and opening a possibility to explore optimal conditions for the readout.

Appendix A

Circuit Quantization

In earlier section, we talked about two important components of a superconducting circuit, quantum harmonic oscillator and artificial transmon atom. Now in this section we'll discuss the interaction between the two components which is crucial for qubit control and measurement. Consider a coupled LC transmon circuit in the Fig(A.1). By using Kirchoff's rule we have,

$$-\dot{\Phi}_A + \dot{\Phi}_B + \dot{\Phi}_C = 0 \quad (\text{A.1})$$

The lagrangian of the above system is

$$L = \frac{C_\Sigma \dot{\Phi}_A^2}{2} + \frac{C_g \dot{\Phi}_B^2}{2} + \frac{C \dot{\Phi}_C^2}{2} + E_J \cos \varphi_C - \frac{\Phi_C^2}{2L} \quad (\text{A.2})$$

Using Eq(A.1) in Eq(A.2) to replace $\dot{\Phi}_B$ then we have

$$L = \frac{C_\Sigma \dot{\Phi}_A^2}{2} + \frac{C_g (\dot{\Phi}_A - \dot{\Phi}_C)^2}{2} + \frac{C \dot{\Phi}_C^2}{2} + E_J \cos \varphi_C - \frac{\Phi_C^2}{2L} \quad (\text{A.3})$$

Now we can write the kinetic energy part as $T = \frac{1}{2} \Phi_i C_{ij} \Phi_j$, where C_{ij} is the capacitance matrix given as

$$C = \begin{pmatrix} C_\Sigma + C_g & -C_g \\ C_g & C_g + C \end{pmatrix}$$

we define the conjugate momentum vector as $Q_i = (Q_A, Q_C)$, hence the hamiltonian is given as [Goldstein 02]

$$H = \frac{1}{2} Q_i C_{ij}^{-1} Q_j + V \quad (\text{A.4})$$

In the limit $C_g \ll C_\Sigma$, the simplified hamiltonian is

$$H \approx \frac{[Q_A + (C_g/C)Q_C]^2}{2C_\Sigma} - E_J \cos \varphi_A + \frac{Q_C^2}{2C} + \frac{\Phi_C^2}{2L} \quad (\text{A.5})$$

Raising $Q_i \rightarrow \hat{Q}_i$ and $\Phi_i \rightarrow \hat{\Phi}_i$ redefining the charge operator as $\hat{n} = \frac{\hat{Q}}{2e}$ and $\hat{n}_r = \frac{(C_g/C)\hat{Q}_C}{2e}$. The second term representing an LC oscillator can be diagonalised as $\hat{H}_{LC} = \hbar \omega_m a_m^\dagger a_m$.

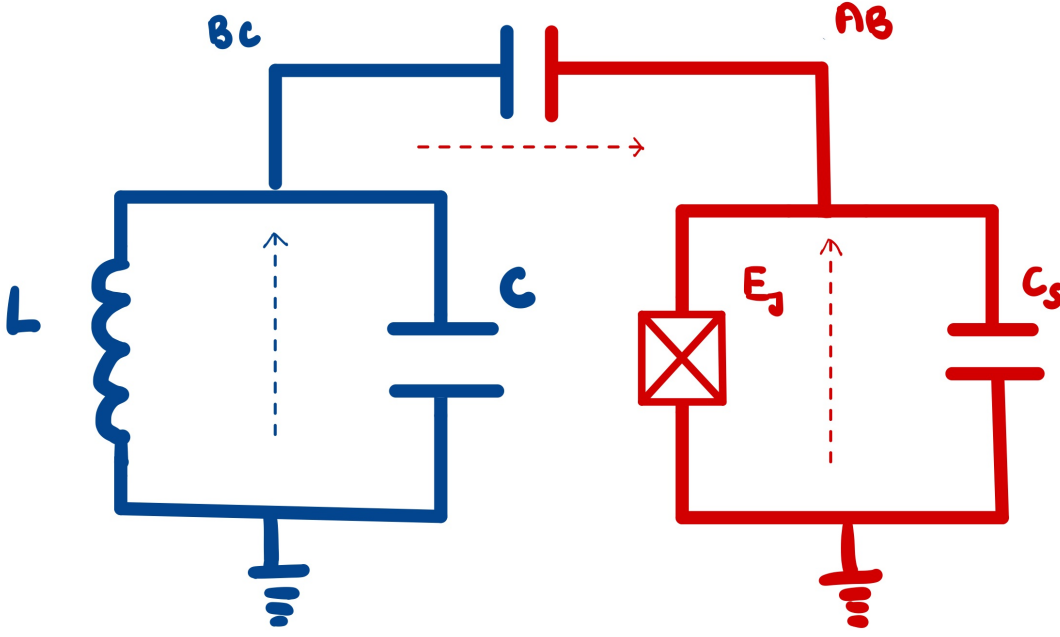


Figure A.1: A Transmon circuit coupled to a LC-oscillator

Hence the hamiltonian in the operator form for a single mode is

$$\hat{H} = \hbar\omega_r \hat{a}^\dagger \hat{a} + 4E_C (\hat{n} + \hat{n}_r)^2 - E_J \cos \hat{\phi} \quad (\text{A.6})$$

Note that Eq(??) is similar to Eq(A.6) with charge-offset $n_g \rightarrow -n_r$ hence explains the coupling can also cause dephasing by modifying the qubit frequency and such effects are more evident at higher levels which are more sensitive to charge noise.

Using creation and annihilation operators introduced in 1.1.2 in single mode approximation we have

$$\hat{H} \approx \hbar\omega_r \hat{a}^\dagger \hat{a} + \hbar\omega_q \hat{b}^\dagger \hat{b} - \frac{E_C}{2} \hat{b}^\dagger \hat{b}^\dagger \hat{b} \hat{b} - \hbar g (\hat{b}^\dagger - \hat{b})(\hat{a}^\dagger - \hat{a}) \quad (\text{A.7})$$

where ω_r is the frequency of the mode of interest and g is the coupling strength between resonator and transmon as

$$g = \omega_r \frac{C_g}{C_\Sigma} \left(\frac{E_J}{2E_C} \right)^{1/4} \sqrt{\frac{\pi Z_r}{R_K}} \quad (\text{A.8})$$

where Z_r is the characteristic impedance of the resonator mode and R_K is the resistance quantum, $R_K = h/e^2$. Using Rotating Wave Approximation (RWA) for $g/\omega_r \ll 1$, in Eq(A.7) we can simplify the hamiltonian as

$$\hat{H} \approx \hbar\omega_r \hat{a}^\dagger \hat{a} + \hbar\omega_q \hat{b}^\dagger \hat{b} - \frac{E_C}{2} \hat{b}^\dagger \hat{b}^\dagger \hat{b} \hat{b} + \hbar g (\hat{b}^\dagger \hat{a} + \hat{b} \hat{a}^\dagger) \quad (\text{A.9})$$

In Eq (A.8), we have term $E_J/2E_C$ which is the size of the vacuum electric field fluctuation and using an appropriate length scale, typically distance moved by the cooper pair

across the transmon's junction. Hence it is tempting to define $\mathcal{E}_0 = (\omega_r/l)(C_g/C_\Sigma)\sqrt{\hbar Z_r/2}$, as the zero-point electric field and the dipole moment given as $d_0 = 2el(E_J/32E_C)^{1/4}$ hence it allows for the coupling to be made much larger than natural atoms.

Appendix B

Transformation Coefficients and Polynomial Numerators

B.1 Third-Order Transformation Coefficients

The raw coefficients for the third-order expansion are given by:

$$\begin{aligned}
 C_k^{(1)} = & -\frac{g_{k,k+1}}{6(\omega_r + \epsilon_k - \epsilon_{k+1})^2} \left[g_{k,k+1}^2 - \frac{(\omega_r + \epsilon_k - \epsilon_{k+1})(\epsilon_{k-1} - 2\epsilon_k + \epsilon_{k+1})g_{k-1,k}^2}{(\omega_r + \epsilon_{k-1} - \epsilon_k)^2} \right] \\
 & - \frac{g_{k,k+1} \left[2\epsilon_{k+1}g_{k-1,k}^2 + \epsilon_{k-1}g_{k,k+1}^2 + \omega_r(g_{k,k+1}^2 - 2g_{k-1,k}^2) - \epsilon_k(g_{k,k+1}^2 + 2g_{k-1,k}^2) \right]}{6(\omega_r + \epsilon_{k-1} - \epsilon_k)(\omega_r + \epsilon_k - \epsilon_{k+1})^2} \\
 & + \frac{(\epsilon_{k-1} - 2\epsilon_k + \epsilon_{k+1})g_{k-1,k}^2 g_{k,k+1}}{2(\omega_r + \epsilon_{k-1} - \epsilon_k)(2\omega_r + \epsilon_{k-1} - \epsilon_{k+1})(\omega_r + \epsilon_k - \epsilon_{k+1})}
 \end{aligned} \tag{B.1}$$

$$\begin{aligned}
 C_k^{(2)} = & \frac{g_{k,k+1}g_{k+1,k+2}g_{k+2,k+3}(\epsilon_k - 3\epsilon_{k+1} + 3\epsilon_{k+2} - \epsilon_{k+3})}{12(\omega_r + \epsilon_k - \epsilon_{k+1})(\omega_r + \epsilon_{k+1} - \epsilon_{k+2})(\omega_r + \epsilon_{k+2} - \epsilon_{k+3})} \\
 & - \frac{g_{k,k+1}g_{k+1,k+2}g_{k+2,k+3}}{2} \left[\frac{\epsilon_k - 2\epsilon_{k+1} + \epsilon_{k+2}}{(\omega_r + \epsilon_k - \epsilon_{k+1})(2\omega_r + \epsilon_k - \epsilon_{k+2})(\omega_r + \epsilon_{k+1} - \epsilon_{k+2})} \right. \\
 & \left. - \frac{\epsilon_{k+1} - 2\epsilon_{k+2} + \epsilon_{k+3}}{(\omega_r + \epsilon_{k+1} - \epsilon_{k+2})(2\omega_r + \epsilon_{k+1} - \epsilon_{k+3})(\omega_r + \epsilon_{k+2} - \epsilon_{k+3})} \right]
 \end{aligned} \tag{B.2}$$

$$\begin{aligned}
C_k^{(3)} = & \frac{g_{k,k+1}}{12} \left[\frac{2(\omega_r + \varepsilon_k - \varepsilon_{k+1})g_{k-1,k}^2}{(\omega_r + \varepsilon_{k-1} - \varepsilon_k)(\omega_r + \varepsilon_k - \varepsilon_{k+1})^2} - \frac{2g_{k,k+1}^2}{(\omega_r + \varepsilon_k - \varepsilon_{k+1})^2} \right. \\
& \left. - \frac{g_{k+1,k+2}^2}{(\omega_r + \varepsilon_{k+1} - \varepsilon_{k+2})^2} + \frac{g_{k+1,k+2}^2}{(\omega_r + \varepsilon_k - \varepsilon_{k+1})(\omega_r + \varepsilon_{k+1} - \varepsilon_{k+2})} \right] \\
& - \frac{g_{k,k+1}}{12(\omega_r + \varepsilon_k - \varepsilon_{k+1})^2} \left[- \frac{(\omega_r + \varepsilon_k - \varepsilon_{k+1})(\varepsilon_{k-1} - 2\varepsilon_k + \varepsilon_{k+1})g_{k-1,k}^2}{(\omega_r + \varepsilon_{k-1} - \varepsilon_k)^2} \right. \\
& \left. + \frac{2(-\varepsilon_{k+2}g_{k,k+1}^2 - \varepsilon_k g_{k+1,k+2}^2 + \omega_r(g_{k,k+1}^2 - g_{k+1,k+2}^2) + \varepsilon_{k+1}(g_{k,k+1}^2 + g_{k+1,k+2}^2))}{\omega_r + \varepsilon_{k+1} - \varepsilon_{k+2}} \right] \\
& + \frac{1}{2} \left[\frac{(\varepsilon_{k-1} - 2\varepsilon_k + \varepsilon_{k+1})g_{k-1,k}^2 g_{k,k+1}}{(\omega_r + \varepsilon_{k-1} - \varepsilon_k)(2\omega_r + \varepsilon_{k-1} - \varepsilon_{k+1})(\omega_r + \varepsilon_k - \varepsilon_{k+1})} \right. \\
& \left. - \frac{(\varepsilon_k - 2\varepsilon_{k+1} + \varepsilon_{k+2})g_{k,k+1}g_{k+1,k+2}^2}{(\omega_r + \varepsilon_k - \varepsilon_{k+1})(2\omega_r + \varepsilon_k - \varepsilon_{k+2})(\omega_r + \varepsilon_{k+1} - \varepsilon_{k+2})} \right]
\end{aligned} \tag{B.3}$$

B.2 Simplified Coefficients via Numerator Polynomials

We can express these coefficients more compactly using auxiliary polynomials:

$$C_k^{(1)} = - \frac{P_{A,k}^{(1)} g_{k-1,k}^2 g_{k,k+1} + P_{B,k}^{(1)} g_{k,k+1}^3}{6(\omega_r + \varepsilon_{k-1} - \varepsilon_k)^2 (2\omega_r + \varepsilon_{k-1} - \varepsilon_{k+1})(\omega_r + \varepsilon_k - \varepsilon_{k+1})^3} \tag{B.4}$$

$$\begin{aligned}
P_{A,k}^{(1)} = & 4\omega_r^3 + 11\omega_r^2 \varepsilon_{k-1} + 6\omega_r \varepsilon_{k-1}^2 - 10\omega_r^2 \varepsilon_k - 2\omega_r \varepsilon_{k-1} \varepsilon_k + 6\varepsilon_{k-1}^2 \varepsilon_k \\
& - 8\omega_r \varepsilon_k^2 - 13\varepsilon_{k-1} \varepsilon_k^2 + 6\varepsilon_k^3 - \omega_r^2 \varepsilon_{k+1} - 10\omega_r \varepsilon_{k-1} \varepsilon_{k+1} - 6\varepsilon_{k-1}^2 \varepsilon_{k+1} \\
& + 18\omega_r \varepsilon_k \varepsilon_{k+1} + 14\varepsilon_{k-1} \varepsilon_k \varepsilon_{k+1} - 5\varepsilon_k^2 \varepsilon_{k+1} - 4\omega_r \varepsilon_{k+1}^2 - \varepsilon_{k-1} \varepsilon_{k+1}^2 \\
& - 2\varepsilon_k \varepsilon_{k+1}^2 + \varepsilon_{k+1}^3
\end{aligned} \tag{B.5}$$

$$\begin{aligned}
P_{B,k}^{(1)} = & -4\omega_r^3 - 10\omega_r^2 \varepsilon_{k-1} - 8\omega_r \varepsilon_{k-1}^2 - 2\varepsilon_{k-1}^3 + 8\omega_r^2 \varepsilon_k + 12\omega_r \varepsilon_{k-1} \varepsilon_k \\
& + 4\varepsilon_{k-1}^2 \varepsilon_k - 4\omega_r \varepsilon_k^2 - 2\varepsilon_{k-1} \varepsilon_k^2 + 2\omega_r^2 \varepsilon_{k+1} + 4\omega_r \varepsilon_{k-1} \varepsilon_{k+1} \\
& + 2\varepsilon_{k-1}^2 \varepsilon_{k+1} - 4\omega_r \varepsilon_k \varepsilon_{k+1} - 4\varepsilon_{k-1} \varepsilon_k \varepsilon_{k+1} + 2\varepsilon_k^2 \varepsilon_{k+1}
\end{aligned} \tag{B.6}$$

$$\begin{aligned}
C_k^{(2)} = & -\frac{1}{\omega_r + \varepsilon_k - \varepsilon_{k+1}} \left(g_{k-1,k}^2 g_{k,k+1} \left(\frac{\varepsilon_{k-1} - 2\varepsilon_k + \varepsilon_{k+1}}{12(\omega_r + \varepsilon_{k-1} - \varepsilon_k)^2 (\omega_r + \varepsilon_k - \varepsilon_{k+1})} \right. \right. \\
& + \frac{\varepsilon_{k-1} - 2\varepsilon_k + \varepsilon_{k+1}}{4(\omega_r + \varepsilon_{k-1} - \varepsilon_k)(\omega_r + \varepsilon_k - \varepsilon_{k+1})(2\omega_r + \varepsilon_{k-1} - \varepsilon_{k+1})} \\
& + \frac{1}{6(\omega_r + \varepsilon_{k-1} - \varepsilon_k)(\omega_r + \varepsilon_k - \varepsilon_{k+1})} \left. \right) - \frac{g_{k,k+1}^3}{3(\omega_r + \varepsilon_k - \varepsilon_{k+1})^2} \\
& + g_{k,k+1} g_{k+1,k+2}^2 \left(\frac{1}{4(\omega_r + \varepsilon_k - \varepsilon_{k+1})(\omega_r + \varepsilon_{k+1} - \varepsilon_{k+2})} \right. \\
& - \frac{1}{12(\omega_r + \varepsilon_{k+1} - \varepsilon_{k+2})^2} \\
& \left. \left. - \frac{\varepsilon_k - 2\varepsilon_{k+1} + \varepsilon_{k+2}}{4(\omega_r + \varepsilon_k - \varepsilon_{k+1})(\omega_r + \varepsilon_{k+1} - \varepsilon_{k+2})(2\omega_r + \varepsilon_k - \varepsilon_{k+2})} \right) \right)
\end{aligned} \tag{B.7}$$

$$\begin{aligned}
C_k^{(3)} = & -\frac{P_k^{(3)} g_{k,k+1} g_{k+1,k+2} g_{k+2,k+3}}{6(\omega_r + \varepsilon_k - \varepsilon_{k+1})(2\omega_r + \varepsilon_k - \varepsilon_{k+2})(\omega_r + \varepsilon_{k+1} - \varepsilon_{k+2})} \\
& \times \frac{1}{(3\omega_r + \varepsilon_k - \varepsilon_{k+3})(2\omega_r + \varepsilon_{k+1} - \varepsilon_{k+3})(\omega_r + \varepsilon_{k+2} - \varepsilon_{k+3})}
\end{aligned} \tag{B.8}$$

$$\begin{aligned}
P_k^{(3)} = & -\omega_r^2 \varepsilon_k + \omega_r \varepsilon_k^2 + 3\omega_r^2 \varepsilon_{k+1} + \omega_r \varepsilon_k \varepsilon_{k+1} + 2\varepsilon_k^2 \varepsilon_{k+1} - 3\omega_r \varepsilon_{k+1}^2 \\
& - 3\varepsilon_k \varepsilon_{k+1}^2 - 3\omega_r^2 \varepsilon_{k+2} - 10\omega_r \varepsilon_k \varepsilon_{k+2} - 3\varepsilon_k^2 \varepsilon_{k+2} + 15\omega_r \varepsilon_{k+1} \varepsilon_{k+2} \\
& + \varepsilon_k \varepsilon_{k+1} \varepsilon_{k+2} + 6\varepsilon_{k+1}^2 \varepsilon_{k+2} - 3\omega_r \varepsilon_{k+2}^2 + 3\varepsilon_k \varepsilon_{k+2}^2 - 6\varepsilon_{k+1} \varepsilon_{k+2}^2 \\
& + \omega_r^2 \varepsilon_{k+3} + 7\omega_r \varepsilon_k \varepsilon_{k+3} + \varepsilon_k^2 \varepsilon_{k+3} - 10\omega_r \varepsilon_{k+1} \varepsilon_{k+3} + \varepsilon_k \varepsilon_{k+1} \varepsilon_{k+3} \\
& - 3\varepsilon_{k+1}^2 \varepsilon_{k+3} + \omega_r \varepsilon_{k+2} \varepsilon_{k+3} - \varepsilon_k \varepsilon_{k+2} \varepsilon_{k+3} - \varepsilon_{k+1} \varepsilon_{k+2} \varepsilon_{k+3} \\
& + 3\varepsilon_{k+2}^2 \varepsilon_{k+3} + \omega_r \varepsilon_{k+3}^2 - \varepsilon_k \varepsilon_{k+3}^2 + 3\varepsilon_{k+1} \varepsilon_{k+3}^2 - 2\varepsilon_{k+2} \varepsilon_{k+3}^2
\end{aligned} \tag{B.9}$$

B.3 Fourth-Order Energy Shift Numerators

$$\begin{aligned}
P_k^{(L)} = & -12\varepsilon_{k-1}^3 g_{k-2,k-1}^2 + 12\varepsilon_{k-1}^2 \varepsilon_k g_{k-2,k-1}^2 + \varepsilon_{k-1} \varepsilon_k^2 g_{k-2,k-1}^2 - \varepsilon_k^3 g_{k-2,k-1}^2 \\
& + 4\varepsilon_{k-2}^3 g_{k-1,k}^2 - 4\varepsilon_{k-1}^2 \varepsilon_k g_{k-1,k}^2 - 8\omega_r^3 (g_{k-2,k-1}^2 - g_{k-1,k}^2) \\
& - 4\omega_r^2 (5\varepsilon_{k-2} - 4\varepsilon_{k-1} - \varepsilon_k) (g_{k-2,k-1}^2 - g_{k-1,k}^2) \\
& + 2\varepsilon_{k-2} \left(\varepsilon_k^2 g_{k-2,k-1}^2 + 2\varepsilon_{k-1}^2 (6g_{k-2,k-1}^2 + g_{k-1,k}^2) + \varepsilon_{k-1} \varepsilon_k (-13g_{k-2,k-1}^2 + 4g_{k-1,k}^2) \right) \\
& + \varepsilon_{k-2}^2 \left(\varepsilon_k (11g_{k-2,k-1}^2 - 4g_{k-1,k}^2) - \varepsilon_{k-1} (11g_{k-2,k-1}^2 + 8g_{k-1,k}^2) \right) \\
& + \omega_r \left(5\varepsilon_k^2 g_{k-2,k-1}^2 + 4\varepsilon_{k-1} \varepsilon_k (-7g_{k-2,k-1}^2 + 2g_{k-1,k}^2) + 4\varepsilon_{k-1}^2 (3g_{k-2,k-1}^2 + 2g_{k-1,k}^2) \right. \\
& + \varepsilon_{k-2}^2 (-11g_{k-2,k-1}^2 + 16g_{k-1,k}^2) + 2\varepsilon_{k-2} (2\varepsilon_{k-1} (g_{k-2,k-1}^2 - 6g_{k-1,k}^2) \\
& \left. + \varepsilon_k (9g_{k-2,k-1}^2 - 4g_{k-1,k}^2)) \right)
\end{aligned}$$

terms $P_k^{(D)}$ and $P_k^{(K)}$ are too complex to be generalized.

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